

**SMARTFLOW: A Simulation-Based Approach to Traffic Signal Optimization
and Adaptive Routing Using Reinforcement Learning**



**A Capstone Project Presented to
Faculty of the Institute of Computing
Davao del Norte State College
New Visayas, Panabo City**

A Capstone Project Presented by:

**Johnfel Anthony B. Caredo
Prille Vincent Salibay
Karen B. Solano**

June 2026

CHAPTER I

INTRODUCTION

Background of the Study

Urbanization, which has led to an influx of people to various cities as well as more people in most cities throughout the world, has brought various problems in traffic management, such as traffic congestion, pollution caused by vehicles, increased fuel consumption among vehicles, and inefficiencies in urban mobility [1]. As the number of vehicles keep increasing, transport infrastructures in many urban areas fail to handle the growing demand for transportation, resulting in longer travel times and economic losses [2]. In this context of various problems facing urban transportation, it is clear that traffic congestion remains a problem to deal with as far as urban transportation is concerned.

Even with the adoption of several methods for managing traffic, a lot of urban centers still depend on the fixed scheme or timed schedule which is also known as the traditional way of functioning of the lights [3]. These traditional methods have little capability to react to changes in conditions. Hence, transportation researchers and authorities have been investigating ways of changing the way traffic is planned and controlled through the intelligent and data-driven systems.

Research has been conducted on using Artificial Intelligence (AI), simulations, and Reinforcement Learning (RL) in intelligent transportation systems to help with traffic signals management and analyzing traffic [5]. Research results reveal that AI-based traffic control can make changes to signals based on simulated traffic flow, while microscopic traffic simulations offer an opportunity to test out different scenarios before implementing any changes [6]. Agent-based simulation allows vehicles and pedestrians to be treated as independent entities, thus permitting a closer look into traffic dynamics [7].

Simulation and AI technologies are enough to show the ability of technology to ensure proactive traffic analysis and decision making.

Amid growing urban mobility pressures in the Philippines, transport systems are consequently overloaded. The Philippine Development Plan (2023–2028) notes the need for information technology-driven approaches to enhance connectivity and efficiency in lines with the use of organized technology aimed at promoting the innovative transport mechanisms. Nonetheless, the deployment of simulation-based decision-support systems is yet to be maximally utilized on a local scale. Therefore, local systems that can take into consideration the existing road condition patterns and offer options in traffic management may give more room for making decisions and further planning.

This study centers on a specific junction and the nearby road network located in Tagum City, Davao del Norte. Validation of the research area will be conducted by means of field observational technique and consultations held with pertinent local government agencies to ensure that the simulated road configuration and traffic conditions reflect corresponding ground conditions. Information dealing with road and traffic conditions in the real sector shall be used as references for simulation development when available from authorized sources.

The proposed system, called SMARTFLOW, is to be used for both evaluating traffic movements in conjunction with the traffic network and as a decision-preparation instrument based upon simulated outcomes and data. This system will enable users to set up the various elements related to traffic and study incident scenarios, such as road closures and other factors that could impact normal traffic operations. This system will also assist in determining various routing schemes and the configuration of traffic signals based upon the simulation's data. One of the purposes of SMARTFLOW is to help simulate the implications of potential actions regarding the management of traffic prior to performing them.

This research contributes to the implementation of the Davao del Norte State College Research, Development and Extension (RDE) Agenda 2026–2030 particularly the Innovation and Inclusion (I2) Agenda of the college by utilizing ICT for a localized transportation issue[19]. The study is also aligned to that of the UN Sustainable Development Goal 11 (Sustainable Cities and Communities) that creates more efficient, resilient and sustainable urban mobility [10]. Moreover, the study is not only synonymous to the CHED A.C.H.I.E.V.E. Agenda, but also its Harmonized SDG-Based Research and Innovation goal which advocates for researches and innovations that respond to local, national and international concerns as well as contributing to the achievement of the Sustainable Development Goals [10], [16]. Lastly, the research supports the Department of Science and Technology Harmonized National Research and Development Agenda (HNRDA) in the sustainability pillar as it employs science, technology and innovation for a local transport and urban mobility matter [3].

However, there is still a strong demand for localized decision-support systems that have the capacity to utilize site-specific data and give the traffic management entities the chance to observe how the road disturbances, routing choices, and traffic light arrangements would likely behave before any changes to the built environment are made. In the case of Tagum City, the ability to validate the results of a simulation through the use of current roadway conditions and traffic data could provide useful information for traffic-related decision-making purposes. The main purpose of this study is to develop a program called SMARTFLOW that will enable localized simulation-based traffic analysis, scenario testing, evaluation of routing methods, and signal planning.

Objectives of the Study

General Objective

To create SMARTFLOW, The main objective of this project is to develop a Reinforcement Learning-based simulation approach for traffic signal optimization and adaptive routing for urban development planning. Specifically, this study aims to:

1. Design an environment of simulation to model a specific real-world intersection in Tagum City which can include road maps, traffic flow statistics, and geographical features around the intersection to be modeled.
2. Use actual traffic data for modeling purposes to simulate the traffic volume and traffic state for the modeled intersection.
3. Create a simulation interface which enables users to set traffic parameters (like size and timing of signal systems) to design and test possible traffic scenarios.
4. Run different types of scenarios on the model (road closures, modification of existing roadways, etc.) to analyze the effect of proposed changes on traffic flow and to develop plans for rerouting vehicles.
5. Conduct modeling of event driven scenarios (e.g. peak traffic times, holidays, road closures, etc.) in order to study traffic behavior under different conditions.
6. Use reinforcement learning technology to allow the traffic signal to learn from the traffic conditions and provide information on how to arrange traffic signals.

Significance of the Study

This study aids off the development of simulation-based and data-driven traffic management by amalgamating traffic data from real-world, traffic simulation, road closure and rerouting scenarios, and Reinforcement Learning-based traffic signal control. SMARTFLOW creates an environment for evaluating traffic conditions as well as possible traffic management strategies prior to implementation. Through the simulation of both existing and proposed traffic signal conditions, the system supports analyzing whether a certain intersection will benefit from traffic signal and which signal configuration can result in better traffic performance.

The findings of this research is likely to serve advantages to:

- **Local Government Units (LGUs) and Traffic Management Authorities.** The system will potentially act as a decision-support tool for assessing the existing traffic situation with respect to the potential installation of signals and development of signal operations, possible closures of roads as a result of construction activity, and alternative routing options. This will help traffic management authorities go through possible traffic management options that could be implemented prior to applying changes to the current traffic situation on the roadway.
- **Traffic Planners and Urban Developers.** This simulation will provide traffic flux, queue configuration, waiting time, and other traffic performance metrics in various scenarios. This will help traffic planning and evaluation with respect to the implementation of signal systems, road management techniques, and other transportation interventions.
- **Traffic Enforcement Officers.** The system can be utilized by traffic enforcement officers to study the potential ramifications of various traffic events on road closures, rerouting, and overall traffic flow within the study area. This will yield more preparedness and coordination for any disruptive traffic conditions.
- **Drivers and Users of the Roadways.** The evaluation of signal installation, signal operation, and alternative routing opportunities will

serve the purpose of enhancing traffic flow therefore reducing delays and enhancing travel experience.

- **Pedestrians.** The simulation will take into account pedestrians and pedestrian movement and activity will be considered when determining the operational characteristics of traffic signal systems and traffic situations.
- **Researchers and Future Developers.** The results of this research can be used as a reference for future studies containing traffic simulation, reinforcement learning-based traffic signal control, traffic rerouting, and event-based traffic scenarios in addition to data-driven decision-support mechanisms for urban transit.

Scope and Limitations

The subject of the present research is SMARTFLOW development, a simulation-based traffic analysis and decision support tool for a given intersection and surrounding road constructions nearby Tagum City in Davao del Norte. The method for simulation creation will be developed using the Simulation of Urban Mobility (SUMO). The simulation site will be created on the basis of the actual road construction of the selected study site. Various data collection methods will be used for validation of the simulation site. The available traffic data which was acquired from appropriate sources located in Tagum City. The traffic data will include traffic volume and relevant traffic conditions within the site simulation. Other landmarks located in the study site and other road condition that were used in simulation site building will also be used in the simulation development.

The simulation denotes vehicles and pedestrians as traffic entities that are created according to traffic volume and conditions specifically being acquired by data provided or accomplished. The component of Reinforcement Learning will be applied in the traffic signal control process, in which traffic signal control is an agent of learning. The agent will observe traffic conditions that are being simulated and will choose an appropriate signal action according to certain performance metrics defined. This will enable evaluation of the existing traffic

conditions and proposed signaling placement and configurations at pre-selected intersection areas that include those areas that do not currently have a signalization.

The SMARTFLOW system will enable the simulation of various traffic scenarios, including normal traffic situations, peak traffic hours, weekends, holidays, road barriers, and accidents, among other cases of traffic disturbances. Furthermore, the SMARTFLOW system will allow for road closure and detour assessment, which will enable the analysis of other road configurations as well as how traffic flows based on the initial configuration of the roads following their closure. Traffic simulation performance measures generated by SMARTFLOW will ultimately include vehicle waiting time, queue counts, traffic volume, travel time, and traffic density, among others.

This research has been conducted on a small area with a specific intersection and some connects junctions and roads. It does not claim to be representative for the whole of the Tagum City. The effectiveness of the simulation process will depend on the accessibility of the relevant traffic data obtained from references and records. As a result, the simulated data will provide estimates of modeled traffic conditions based on available values without giving actual measurements on an instantaneous traffic basis.

The system does not include live traffic feeds and CCTV, GPS tracking, IoT devices, and real-time traffic sensors will not be part of the system. The technology will also not interface with or control any traffic signals and/or the actual physical traffic infrastructure. Therefore, the suggested use of traffic signals will only be judged in a simulation environment and will not be seen as an actual recommendation for implementation in the field until further engineering evaluations and approvals are made.

The Reinforcement Learning component will only focus on traffic signal operation in the simulated environment as it affects its results. Quality of traffic data, simulation settings, defined traffic scenarios, and rewards and performance

metrics set by the contributors will determine the effectiveness of this component. This research does not include construction, installations, or operations of traffic signals, physical modifications to roads, or implementation of any rerouting schemes on the actual road network.

Review of Related Literature and Studies

Related Literature

Traditional Traffic Management Systems

The conventional approach to managing traffic remains based on fixed signal schedules and manual enforcement of traffic regulations [11]. Fixed signal control involves pre-determined settings derived from the previous research done on the traffic volume [12]. Although the fixed signal control is convenient and simple to operate, its predetermined signal timing is often not adaptive to traffic fluctuating demand led to queuing and excessive delays [13]. The research undertaken on the smart control of the intersections suggests that the applied computer-based technologies make it more efficient and save time compared to the other traffic management strategies [14]. The limitations discussed above suggest the need for conducting the assessment of the traffic conditions as well as evaluating several possible signal configuration options prior to the actual implementation.

Intelligent Transportation Systems (ITS) in the Philippine Context

Intelligent Transportation Systems (ITS) are a combination of information technology, transportation data, and other data-driven solutions in order to manage traffic flow, enhance mobility, and promote road safety [15]. In the context of the Philippines, the development of ITS implies going hand in hand with the smart-city movement, in use of the technologies to solve urban transportation problems [16]. As previous research showed, the adaptive traffic management approaches are utilized through traffic data to adjust the traffic management systems to the changes in traffic conditions [17]. The existing ITS

technologies can be either based upon real-time sensors or utilizing the existing data in order to simulate efficient ITS applications.

Integration of Artificial Intelligence in Traffic Management

The use of Artificial Intelligence (AI) in traffic control systems has enabled techniques of traffic monitoring and support for transport-related decisions [18]. There have been methods based on AI used for traffic signal control, including procedures applying Deep Reinforcement Learning to find suitable traffic signal operations based on the current traffic situation [19]. In addition to that, the researches have included the use of the method of combined agent reinforcement learning, providing evidence of the applicability of the learning-based part for the control of traffic flow rather than using the traditional systems based on fixed rules [21]. Nevertheless, the aforementioned studies have implemented the use of current data as well as the Internet of Things or several intelligent agents, which is different from the method used in SMARTFLOW. Thus, SMARTFLOW is based on the principle of applying AI specifically through Reinforcement Learning for traffic signal control in a simulated setting with vehicles and pedestrians represented by traffic entities, not autonomous AI agents.

Reinforcement Learning in Traffic Signal Control

Reinforcement Learning (RL) is a machine learning method that allows an agent to gain knowledge about making decisions through its interaction with the environment. It receives feedback on its actions. For instance, in traffic signal control, the traffic signal control is the learning agent while the traffic signals of the simulation make the environment [20]. Earlier works utilizing DL techniques in traffic signal have established the possibility of enhancing traffic performance through simulation [20][22]. RL traffic signal systems use data such as vehicle count and density, waiting time, and queue length as states while phase change of signals to be chosen as actions and traffic performance measures are used as

reward [21][23]. These methods enable the implementation of the RL in SMARTFLOW, where traffic signal control has been used as the learning agent and various traffic conditions and traffic signal control configurations have been evaluated in addition to evaluating the possible implementation of the traffic signal at a specific intersection.

Agent-Based Traffic Simulation and SUMO

Agent-based and microscopic traffic simulation methods provide a very detailed representation of the individual traffic entities and their interactions in a road network [24]. These methods allow for the representation of vehicles and pedestrians separately in a simulated environment; however, they do not operate in the same way as autonomous agents with artificial intelligence. The Simulation of Urban Mobility (SUMO) is an open-source microscopic and multimodal traffic simulation tool, which allows for modeling vehicles, pedestrians, road networks and traffic signal operation. Furthermore, SUMO incorporates the TraCI interface which makes it possible for external applications to control and interact with simulation elements while it is running [25]. Earlier studies used SUMO for examining traffic signal control and use of Reinforcement Learning [25]. In SMARTFLOW, SUMO will be used to simulate the selected intersection as well as connected road segments in the city of Tagum using the validated road design and local traffic data. The traffic signal controller will be used to interact with the simulation as Reinforcement Learning agent for the implementation of the project.

Traffic Data-Based Simulation

The practical implementation of traffic simulation is strongly associated with the quality of input data regarding the genesis of traffic generation for the simulation model. Traffic input data with respect to traffic flows, traffic counts, traffic generation rates, and other relevant traffic information can be used as inputs for traffic generation for the simulation model [26]. Use of observed or officially published traffic information can prove to offer a more realistic simulation

compared to hypothetical traffic levels. In this study, local traffic data available for Tagum City will be utilized to estimate the traffic inputs to be used within the simulation environment for the study area. Field verifications and consultations with relevant local government offices would also be performed in order to validate if the road configurations and characteristics of the specific intersection selected remained consistent.

Road Closure and Traffic Rerouting

Various incidents such as accidents, construction, maintenance, emergencies, public events and other disruptions lead to closings of roads which in turn may throttle traffic flow by limiting the capacity of the roads and making vehicles to divert to other roads [27]. With the use of traffic simulation on road closures, the potential effects of road closing as well as options for diversion can be known even before any such restriction takes place [28]. A section of road in question which is restricted or limited in a simulated environment allows comparison of alternative routes based on travel time, waiting time, queue length and traffic flow. The project will factor in road closures and rerouting in the selected Tagum City road network to analyze how road closure impacts traffic conditions in case of road closure so as to discover alternatives.

Event-Based Traffic Simulation

Traffic situations change based on the intensity of traffic at certain times, traffic conditions on roads, availability of road usage, and external events. For instance, peak times, weekends, public holidays, collisions, road constructions, and public ceremonies may create unique traffic situations compared to regular road usage [29]. Event-based or scenario-driven traffic simulation allows testing the situation under controlled conditions and comparing the results [30]. In this regard, SMARTFLOW will implement event-based scenarios in order to test the different situations concerning traffic volume and conditions on the roads, and it will cover scenarios connected to the peak periods, weekends, public holidays, road blocks, accidents, and roadworks.

Simulation-Based Decision Support for Traffic Signal Planning

Simulation models for decision-support systems offer a tool for the evaluation of different strategies in the field of transportation prior to their deployment in the actual road system [31]. Within the field of traffic planning, it is possible to utilize simulation in the comparison of traffic conditions so as to assess the impact of various traffic control measures with parameters such as waiting time, queue length, travel time and volume of traffic. Thus, when dealing with traffic signal analysis the activities require to evaluate not only how to operate an already-existing signal but also establish whether the traffic signal can be helpful at the analyzed point and what specific settings should be used with respect to specific traffic conditions existing there. In this regard, SMARTFLOW is based on the same principle whereby the application of local data, validated Tagum City road environment, event-based scenarios, road closures and diversion procedures and the Reinforcement Learning-based control system are combined together to obtain simulated performance results suitable for the evaluation of possible traffic point placement.

Related Studies and Systems

Dual-Targeting Deep Reinforcement Learning Traffic Signal Control System

In the research conducted by Kodama et al. [32], a multi-agent traffic light control mechanism based on a dual-targeting approach of deep reinforcement learning (DRL) in conjunction with Simulation of Urban Mobility (SUMO) was suggested. The proposed model works with traffic-state parameters like the positions of vehicles, density of the lanes, the variations found in density, and the speed of the vehicles, so that traffic signal commands can be prescribed while minimizing the total wait time from a number of vehicles as a whole. Each of the

situations under deliberation shows how the SUMO platform can provide a good environment for training a traffic signal controller based on reinforcement learning. Moreover, such a piece of research is directly connected to SMARTFLOW, given that both models apply reinforcement learning and SUMO-based approach for traffic signal controlling; whereas the only focus of the study was on identifying the optimization meaning of traffic signals with the help of DRL, SMARTFLOW broadens the entire idea associated with the use of simulation by adding a validated localized intersection in Tagum City, real data on traffic and occurrences based on traffic events, and management of traffic such as closure and rerouting.

The framework consists of a neural network handled by a Traffic Signal Control System (TSCS). As illustrated in the conceptual diagram in Fig. 1, the input state includes vehicle position (P_t), lane density (D_t^l), change in density (ΔD_t^l), and velocity (V_t^l), which are processed through multiple hidden layers (256 and 128 ReLU units) to output Q-values for action selection. The system operates through a closed-loop interaction where the SUMO environment provides traffic states, and the RL agent issues signal commands (actions) while receiving rewards based on the minimization of cumulative waiting time.

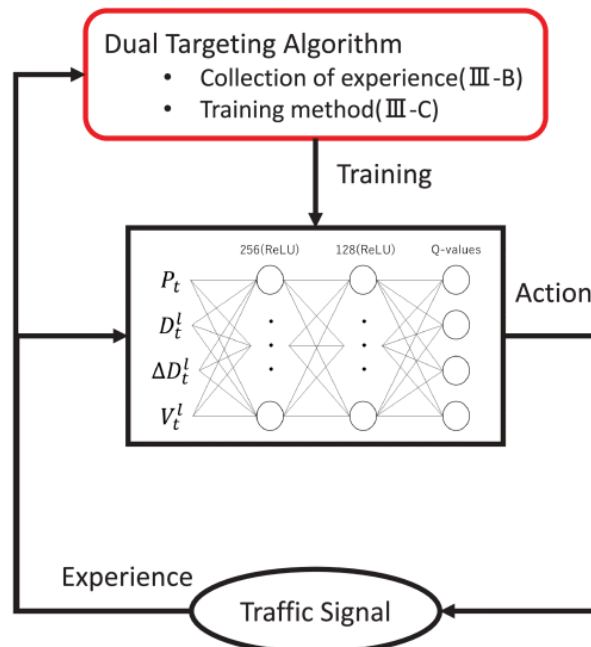


Fig. 1. Conceptual diagram of the proposed traffic light control system.

Extensive simulations were performed to test the effectiveness of the dual-targeting algorithm. The results of these simulations are shown in Fig. 2. Here, learning curves for each of 1000 trials are shown. Although both the standard DQN and Multistep DQN architectures exhibited greater variance, the proposed “Dual-Targeting” method provided a much lower unbiased sample variance of 0.02 after 1000 trials. Such evidence from the simulations validates that the agent achieves stabilization and reduction in vehicle waiting time as the agent learns within the simulation environment.

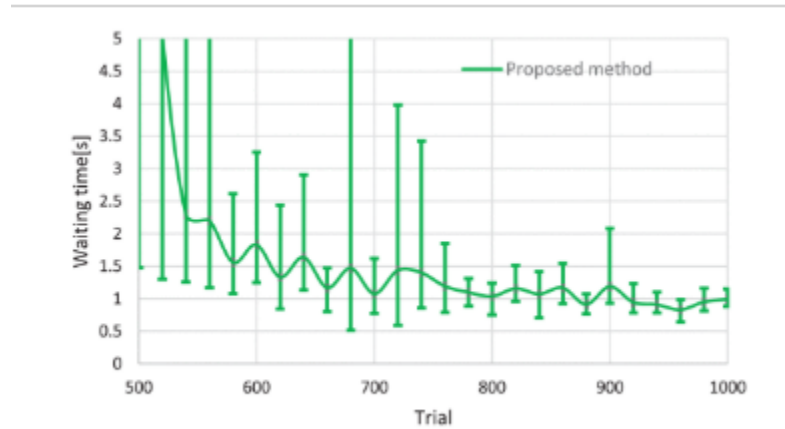


Fig. 2. Simulation results showing the reduction in waiting time using the proposed RL method.

The system is closely related to SMARTFLOW since both make use of Deep Reinforcement Learning (DRL) and the SUMO environment to optimize

traffic signals. It explains the techniques to determine traffic state and how to design reward function from traffic related parameters, e.g. total waiting time [30]. However, SMARTFLOW is more advanced in that it uses these methods on an intersection in Tagum City which has already been verified to be similar to real-life circumstance.

TrafficSim: Learning to Simulate Realistic Multi-Agent Behaviors

According to a research study by Suo et al. [33], the TrafficSim introduced the Multi Agent Behavior Model which can provide realistic and socially realistic traffic simulations in the area of real life data. The TrafficSim study creates traffic behavior through human demonstration and models the behavior of traffic participants through neural network architecture. This study opens avenues for data-driven simulations of traffic as well as producing realistic traffic behavior, even though it cannot be implemented for SMARTFLOW due to the study's difference. On the contrary to TrafficSim, SMARTFLOW will make use of SUMO to portray vehicles and pedestrians as traffic entities through the application of the local traffic data, while Reinforcement Learning will be implemented on the traffic signal controller. Hence, while TrafficSim has shown evidence for providing data-driven traffic simulation, but SMARTFLOW will utilize localized traffic signal evaluation related to traffic management circumstances.

TrafficSim makes use of an implicit latent variable model to generate socially consistent plans for each agent in a scene. The architectural framework of TrafficSim is presented in Fig. 3. Here the system has taken the bird's eye view (BEV) of the map, which includes the states of agents into account through a CNN based encoder. It then employs a graph neural network (GNN) in order to model the interactions amongst the various agents on the scene. Finally, it uses the MLP decoder to predict future trajectories or movements of agents over a longer time duration.



Fig. 3. System architecture of TrafficSim illustrating the CNN encoder, interaction graph, and trajectory decoder.

The effectiveness of the TrafficSim system has been illustrated in the form of its ability to create diverse and socially consistent trajectories for multiple agents in a multi-agent model. An illustration can be seen in Figure 4 where for each vehicle agent, multiple feasible future trajectories have been predicted, using a single observation of the given environment. When the predictions, marked in red and blue are compared to the actual trajectory, predicted as green, it highlights the fact that the TrafficSim system has made accurate predictions about the behaviour of drivers in normal conditions, including the driving behaviour characteristics of maintaining lane and avoiding collision, essential for the realistic urban modeling.

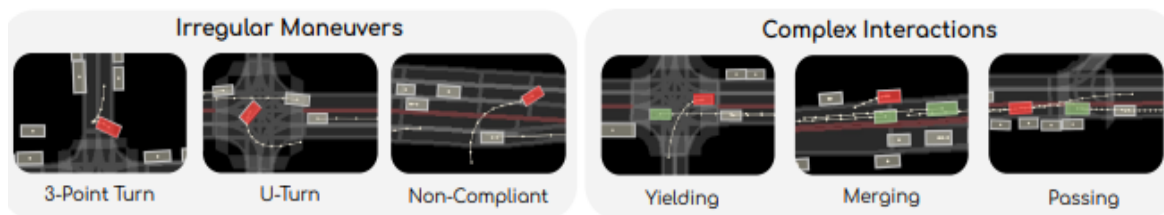


Fig. 4. Visual output of TrafficSim showing diverse multi-agent trajectory predictions and ground truth comparison.

The relation of this study to SMARTFLOW is because it shows how actual data and data-based traffic simulation can be used to model traffic behavior in a simulated setting. TrafficSim will deal with learning realistic vehicle trajectories for self-driving tests while SMARTFLOW will focus on modeling vehicle demand in terms of traffic data at an intersection of Tagum City. The difference here is that TrafficSim models autonomous learning-agents for the vehicles while

SMARTFLOW will use reinforcement learning to add to the optimization of the traffic signal controller and will test the various signal strategies within the simulation environment.

IoT-Based Adaptive Traffic Signal Controller for Congestion Reduction

AlMulla et al. described an adaptive traffic light controller designed based on Internet of Things (IoT) technology and utilizing ultrasonic devices in detecting the presence of vehicles and their queues to change the timing of the traffic light signal. The experiment shows the way traffic-related information is utilized for the modification of the signals changing cycles and the reduction of waiting time of cars as a result, as compared to the standard system that is static. Although the experiment is directly related to SMARTFLOW because it demonstrates the effectiveness and usability of traffic-controlled signal systems, their methodology is based on the use of physical sensors and real-time systems. Whereas SMARTFLOW is based only on the simulation system using SUMO simulation environment and available traffic information and data, comparing the systems used. SMARTFLOW has the unique advantage of allowing the user to analyze such factors as the roads configuration and changes, in case of emergencies.

The architecture of this system uses an Internet of Things (IoT) framework that incorporates the placement of ultrasonic sensors at fixed spacing along the roadway to monitor the presence of vehicles and measure the length of the queue. According to the architecture shown in Fig. 5, the information obtained from the sensors is sent to a central controller that uses a smart algorithm to modify the default green time of 60 seconds. The controller uses the level of congestion it detects to dynamically either increase or decrease the amount of time allocated for each phase of the traffic signal, thus ensuring that the junction is able to adapt flexible traffic conditions.

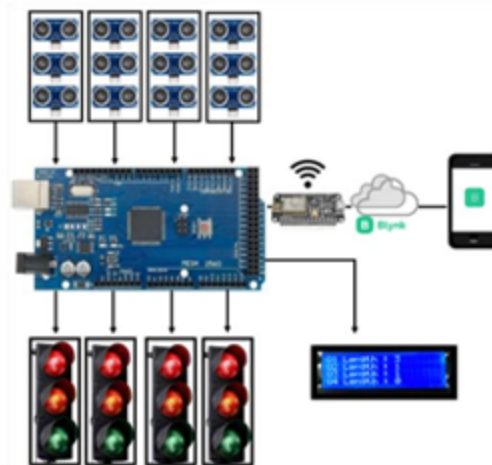


Fig. 5. System architecture of the IoT-based adaptive traffic signal controller.

Performance of the system was validated with a working prototype, along with a mobile app. The system indicates congestion levels like low, medium and high as shown in Fig. 6 using smartphone interface. The study showed that adaptive method increased the overall traffic flow and also reduced the average waiting time significantly at the intersection.

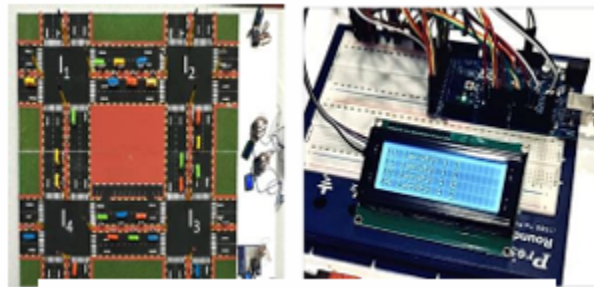


Fig. 6. Sample output showing the real-time congestion monitoring application.

The relevance of this study to SMARTFLOW is shown by the practical use of traffic data such as the queue length in an adaptive traffic signal controls

system. Similar to the work of AIMulla et al., SMARTFLOW also uses traffic conditions as a basis for the evaluation of traffic signals. However, unlike the present system that utilizes physical IoT sensors for real-time traffic detection, SMARTFLOW does employ SUMO and existing traffic data to develop localized traffic conditions. This system allows SMARTFLOW to test the signal configurations, road closures, changes in routes, and many more events without making any changes to the physical sensor system.

Synthesis of Related Studies and Systems

The studies reviewed have shown Reinforcement Learning, traffic simulation, and adaptive signal control can contribute significantly to traffic management. For example, Kodama et al, used Deep Reinforcement Learning and SUMO for traffic control while Suo et al, used real-world data for detailed traffic simulation. AIMulla et al showed the use of traffic information for adaptive control based on IoTbased physical systems. Thus, these studies mention important approaches for traffic signal optimization, data-based simulation and traffic-responsive control activities.

Nonetheless, the systems analyzed above are not in accordance to the intended purpose of SMARTFLOW. Existing works have concentrated on adaptive traffic signal control, data-driven traffic simulation or real time sensor-based traffic management while not putting emphasis on utilizing localized traffic data and a proven road environment to determine proposed traffic signal placements and configurations for various traffic scenarios. Further, very few cases integrate road closure and rerouting approaches with Reinforcement Learning based evaluation of traffic signals in a localized Philippine intersection.

To fill the existing gap, SMARTFLOW has developed a simulation-based traffic decision-making support system targeted at a particular location including Tagum City's main intersection as well as relevant road sections. The model for the present project will incorporate the characteristics of the considered locations

including updated information collected locally to simulate traffic flow within SUMO. The traffic signal controller will be improved by using Reinforcement Learning for the valuation of the selected signal configuration and actions. Event-based methods will be used to enable discussions of emergency road closures and rerouting of vehicles from one point to another in order to investigate the impact of different traffic scenarios on the traffic flow for the studied area.

Table 1.Comparative Analysis of Related Traffic Management Systems and SMARTFLOW

Features	Dual-Targeting DRL [32]	TrafficSim [33]	IoT-Based Adaptive Traffic Signal Controller [34]	SMARTFLOW
Main Technology	Deep Reinforcement Learning	Multi-Agent Trajectory Learning	IoT + Adaptive Signal Control	RL + Agent-Base Simulation
Simulation Environment	SUMO	Learned Traffic Environment	Prototype-Based Traffic System	SUMO
Adaptive Signal Control	✓		✓	✓
Traffic Data-Based Simulation		✓		✓
Autonomous		✓		

Agents				
Road Closure Analysis				✓
Rerouting Analysis				✓
Event-Baseed Scenarios				✓
Localized Traffic Simulation				✓
Traffic Signal Placements Evaluation				✓
Real-time Infrastructure Dependency			✓	
Decision Support Function				✓

Table 1 summarizes the analysis of the literature on traffic management related to the proposed SMARTFLOW system. In contrast to the other models analyzed such as the use of deep reinforcement learning, traffic simulation using data-driven multi-agent systems, and the use of IoT-based signal control systems in traffic management as presented below, where a large part of the focus is on implementation and differs from one study to another. The Dual-Targeting DRL model demonstrates the effectiveness of the SUMO and RL for adaptive signal

control, while on the other hand the TrafficSim model employs a method of learning realistic traffic patterns from data. The use of the IoT-based system is able to provide real time traffic signal control while utilizing information based on pre-constructed infrastructure. The SMARTFLOW system utilizes a local, simulation-based approach utilizing available traffic data and a proven Tagum City road environment as follows. The use of reinforcement learning applied to traffic light signals, analysis of road closure and rerouting, event -based scenarios, and evaluation of possible traffic signal installations are all included in the SMARTFLOW model.

Definition of Terms

Agent-Based Simulation. A type of simulation modeling approach that uses models in which individual entities (e.g. vehicles, pedestrians), each of which can act as an agent with individualized autonomy to allow for interactions among them using technologies in simulations.

Artificial Intelligence (AI). A definition of AI is that the term refers to the capability of computer systems to accomplish tasks that normally require the use of human intelligence to carry out the activity, such as learning how to complete tasks by itself.

Autonomous Agent. Autonomous agents are independent entities capable of interacting with their environment, making decisions and/or carrying out actions without the need for human intervention.

Decision-Support System. A type of system used by computer systems to assist users in analyzing and making decisions about activities based on an analysis of scenarios derived from simulated events and simulations through data analysis processes.

Emergency Vehicle Priority. A traffic management process where the timing of traffic signals is changed to allow for equipment to gain faster entry into the area where it is needed via the operation of traffic signals.

Heterogeneous Agent Behavior. Differences in behavioral patterns associated with agents used in simulations to depict daily activities performed by agents,

including different characteristics for different agents based on their behavior as drivers and in terms of walking behavior.

Intersection Traffic Flow. Refers specifically to traffic causing a movement and interaction of any agent type at an intersection controlled by a traffic signal.

Queue Length. A measure of the number of vehicles waiting at the same signalized intersection.

Reinforcement Learning (RL). A type of machine learning used to develop optimal actions by having the agent observe its environment and receiving rewards or penalties for its actions based on how well its performed.

Road Constraint. A definition of some condition that prevents the normal flow of traffic across the roadway (e.g., lane closure, road construction accident).

Simulation of Urban Mobility (SUMO). An open-source "traffic simulation software" used to model traffic movement based on the actions taken by individuals using the software.

Traffic Signal Control. The process of managing actions within the traffic signal process to control the flow of traffic at the intersection.

Traffic Simulation. A type of traffic simulation where computer simulation can be used to model the flow of traffic in the virtual system.

Adaptive Traffic System. An automated traffic management system that can change the timing of traffic signal systems as traffic conditions and traffic demands change.

Pedestrian Demand. The number of pedestrians who need to cross a certain street or intersection.

Traffic Density. The density of traffic in terms of the number of vehicles present on a specific roadway at any point in time.

CHAPTER II

METHODOLOGY

This work employs the Hybrid Development Methodology, the complementarity between the structured procedures of the Waterfall model and the flexible strategies of Agile. In the first stages of the project, the Waterfall model served as a base for the study including planning, requirements elicitation, and system design. This way, precise objectives in the form of system specification and requirements for traffic simulation and traffic modeling were achieved before the development process began. In contrast, the Agile principles were applied at the stages of implementation, testing, and refinement. Thus, iterated simulation, continuous testing, debugging and repeated improvement of the simulation environment, traffic signal controller based on reinforcement learning (in particular the developed dashboard) could be performed. Therefore, the combination of the Waterfall approach and Agile principles helped create a balance of structure and flexibility, which perfectly matches the purpose of AI-based traffic simulation.

The project starts with the Planning Phase where the project team's members define the purpose, goals, timetable, and resources which will be needed based on what was outlined in the approved project proposal. In this phase, they also determine the study site and traffic scenarios to use in the simulation and define the viability and direction of the system.

During the Analysis Phase, the project team collects final information regarding the system requirements through the study of traffic condition, traffic pattern, pedestrian behavior, and requirement details to develop the simulation and reinforcement learning applications.

Within the Design Phase, the promoters turn all identified requirements into organized architecture of the system, which includes all designs of the workflow of simulation, framework of reinforcement learning, dashboard interface, and system interactivity flow. Also, the developers create several

diagrams and models of the system, so as to provide a visual interface of the operation of the system.

During the implementation phase of the project, the SMARTFLOW system will be developed using an iterative approach. Development cycles of simulation environment and reinforcement learning integration and dashboard interface will be carried out. Development cycles will involve building, testing, and improvement of the system components for various iterations. The implementation will involve configuration of SUMO simulation environment, development of agents for both vehicles and pedestrians, integration of the Python and TraCI communication protocol, design and development of reinforcement learning model, and local dashboard and SQLite database. The key objective of this phase will be to prepare the core components of the system for the final testing and evaluation.

The proponents carry out several series of tests during the Testing Phase of the project to assess how the system behaves with regards to various traffic scenarios and road constraints. The proponents take into account the simulation environment and traffic lights response, the results they obtain from reinforcement learning, and the display of data on the dashboard.

The Refinement Phase is marked by the refinement of the system which involves fixing all detected problems and making adjustments to the simulation processes. It represents the last phase of the project. Although the refinement of the system is only performed for the duration of this particular study, it is expected that this will help maintain the stability of the system during future tests.

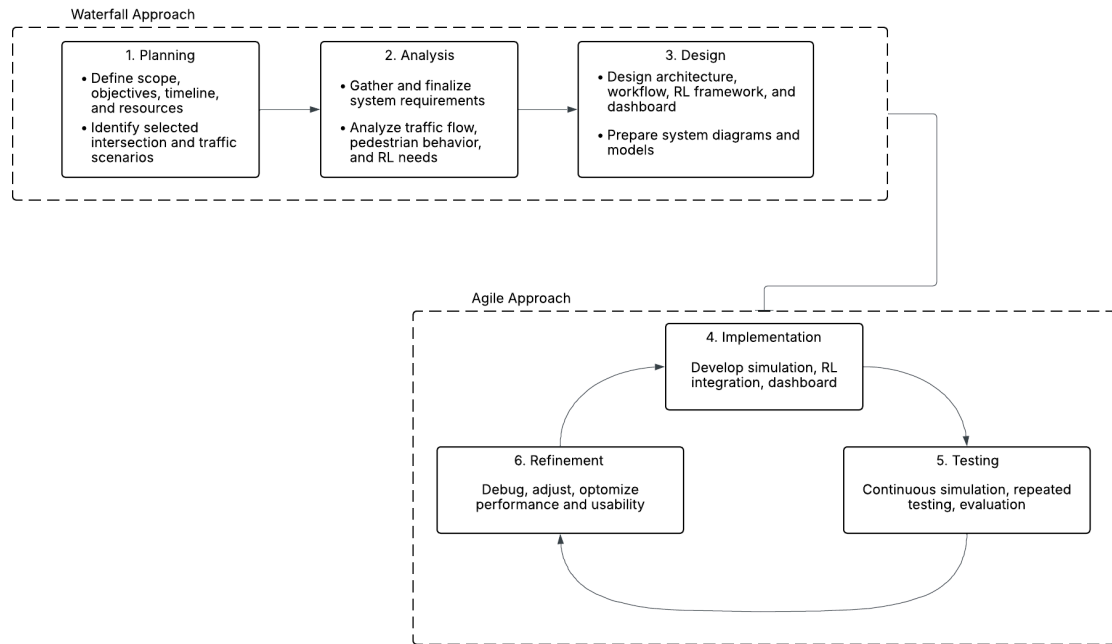


Figure 7. Hybrid Development Methodology

SYSTEM PLANNING

Project Team Organization

As can be noted from Figure 8, the project team is structured in such a way that allows for effective coordination, good communication, and appropriate delegation of responsibilities during the process of developing the SMARTFLOW system. The Advisor provides the academic as well as the technical advice needed to ensure that the study adheres to the established standards of research. The project manager and the documents leader manages the planning phase of the project, provides assistance with coordination of tasks, monitors progress, supervises the scheduling of activities, and manages the creation of relevant project documents. Under this role is an individual known as the system developer and the system analyst. The system developer has the responsibility of programming, implementing the system, developing simulations, and integrating the technology. The system analyst has the responsibility of analyzing

the requirements of the system, discovering the various processes of the system, and assisting in the formulation and evaluation of the SMARTFLOW system. Such an organization makes it possible for the team to have coordinated efforts in providing academic guidance, project management, documentation, technical development, and system analysis.

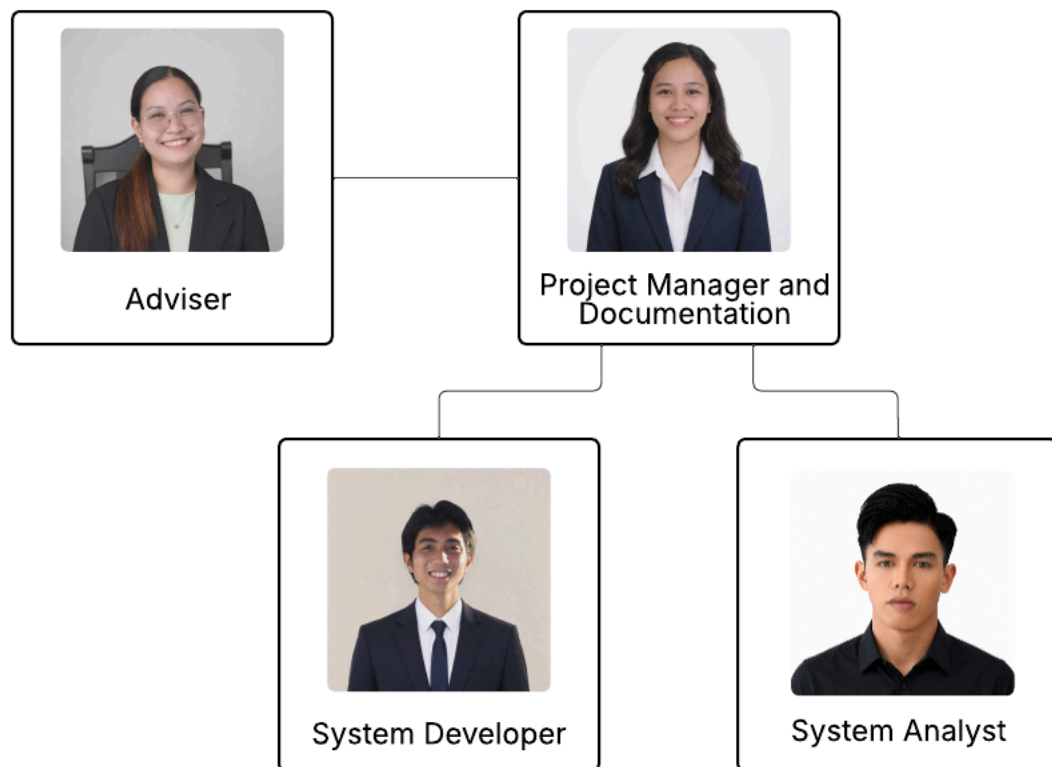


Figure 8. Project Team Organization

Work Breakdown Structure

The display of the Work Breakdown Structure in Figure 9 shows how the SMARTFLOW Traffic System Development is broken down on the basis of its phases. This methodology for the development of the project is categorized into

two parts (the Hybrid Development Method) in which the earlier phases use the Waterfall Methodology while the latter ones use Agile approach. There are 6 phases in total in this WBS.

The phases include, Planning, Analysis, Design, Implementation, Testing and Evaluation, and Refinement. In the Planning Phase, the authors will discuss the scope of the project and its purpose, create a timetable for the project, specify the location for the selected traffic incidence location in Tagum City, and describe the various traffic instances. The phase also serves to lay the groundwork for the development phase by clarifying the goal of the project and its content needs to conform to any traffic challenges that should occur during the entire development of the system.

The Analysis Phase is about understanding the traffic situation and the requirements of the system. This involves examining the various types of traffic flow behavior, the pedestrians' movements and the requirements for the simulation and reinforcement learning components. The proponents also identify the necessary functional and non-functional requirements of the system.

The Design Phase is where the structure and models of the SMARTFLOW system are prepared. This involves the design of the system architecture, simulation workflow, reinforcement learning framework, dashboard, and system diagrams. These designs provide a guide to how the various components of the system will interact during development and simulation execution.

This Implementation Phase of the project consists of the building of SMARTFLOW, employing the iterative process through its simulation, reinforcement learning, and dashboard. Both the simulation and the integration of reinforcements are going to be developed and improved over time. The proponents will work on the configuration of the SUMO simulation process, the modeling of vehicles and pedestrian agents, the implementation of the reinforcement learning based on Python and TraCI communication, etc. This phase will still concentrate on the implementation stages, as the system has not

been launched yet. This part of the project is going to emphasize the preparation of the core elements that will be necessary during the future tests. Every iteration of the Implementation Phase provides an opportunity to discover the problems and find ways to resolve them, thus improving SMARTFLOW.

The Testing and Evaluation Phase includes the conducting of testing of traffic and road constraints for assessing the system performance. The testing will involve both fixed-time and SMART-based signal controls in order to measure the effectiveness of the project, as well as evaluation of multiple metrics resulting from the implementation process.

The Refinement Stage concentrates on enhancing the system based on the outcomes of the testing and evaluation processes. This involves debugging problems with the system, adjusting the simulation parameters, and enhancing the performance of the reinforcement learning algorithm. The stage concludes with the finalization of the system's output and reports.

In conclusion, the Work Breakdown Structure is a valid plan for the development of SMARTFLOW Traffic System since it organizes the tasks according to the particular purpose of the study.

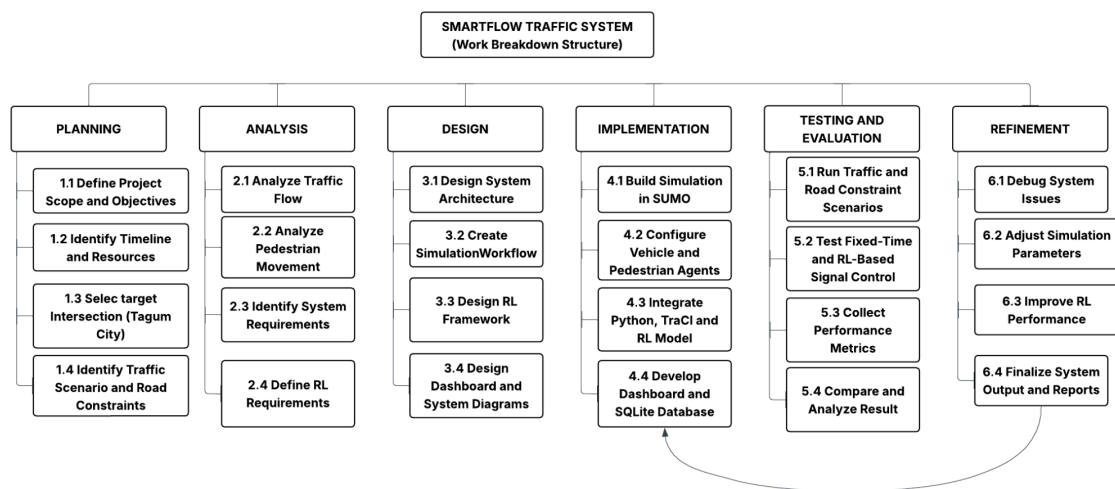


Figure 9. Work Breakdown Structure

Gantt Chart

The Gantt chart depicted in Figure 10 illustrates the timeline and work schedule of the SMARTFLOW Traffic System. The schedule conforms to both the Hybrid Development Methodology and the Work Breakdown Structure adopted in this study. The Gantt Chart begins with the four Waterfall phases (i.e., Planning, Analysis, Design) and ends with the Agile phases of Implementation, Testing and Evaluation, and Refinement.

The Planning Phase consists of determining the objectives and scope of the system project, establishing the time and resources required, selecting an intersection for the project, and determining the parameters of the project and constraints of the roadway under consideration. During the Analysis phase of this project, problems of traffic flow, pedestrian flow, the needs of the system, and the needs of reinforcement learning are examined. The Design phase consists of creating the system architecture, simulation process, environment with reinforcement learning, dashboards, and models of the system.

The phases of Implementation, Testing and Evaluation, and Refinement are shown as overlapping, sequential, and iterative processes. Implementation will involve creation of the SUMO simulation environment, modeling of vehicles and pedestrians, integration of Python, TraCI and reinforcement learning, and creation of a dashboard and SQLite database. Testing and Evaluation involves running a number of test cases for road conditions, testing the two approaches of traffic signal control, getting traffic performance measures, making comparisons of the results. Finally, Refinement will involve fixing bugs, changing the simulation profile, enhancing the learning capability, and producing final outputs and reports.

The chart shows that the iterative phases will undergo repeated loops based on the results of Testing and Evaluation and improvements in the system.

This is in line with the Agile aspect of the methodology, whereby any problem uncovered in Testing and Evaluation can either be dealt with through Refinement or return to Implementation for correction or enhancement. The chart clearly demonstrates the organizational and time efficient processes involved in developing the SMARTFLOW Traffic System, while still adhering to the six major phases of Work Breakdown Structure.

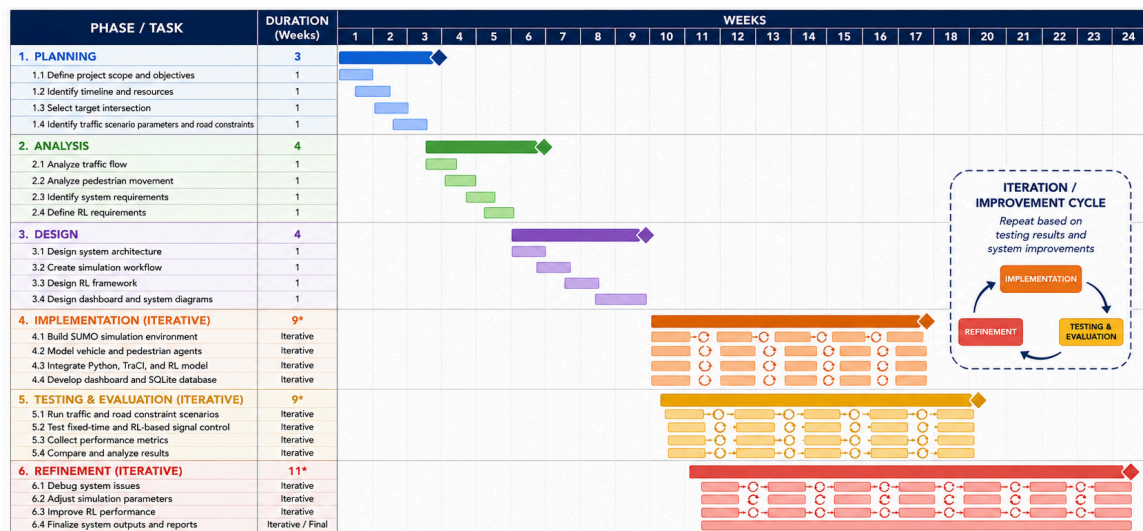


Figure 10. Gantt Chart

SYSTEM ANALYSIS

System Architecture

A broad overview of the architecture of the SMARTFLOW Traffic System is shown in the Figure 11. In fact, the architecture demonstrates the nature of how all components of the system will interact with each other to perform, analyze, and store information about traffic data in the system, using principles of reinforcement learning. The architecture includes the following six layers: the User Layer; the Simulation Layer; the Control and Communication Layer; the Intelligence Layer; the Database Layer; and the Outputs Layer.

The User Layer consists of an authorized operator such as a researcher or a CTTMO employee, who will be able to use the system through a local dashboard. The User Layer allows the operator to log in to the system, simulate traffic scenario, set constraints on roads, select operating mode, start or stop the simulation, monitor the traffic data in real time, and log out of the system.

The Simulation Layer uses SUMO technology for simulation of the localized traffic environment that includes the simulation of roads, vehicles, pedestrians and road constraints. The major aspects covered by this layer are its representation of intersections and lanes, traffic signals, vehicle movements as well as pedestrian crossings and impact of all possible events taking place at the locality like lane closure, accidents, construction etc.

The Control and Communication Layer is the communication link between the simulation environment and the intelligent decision-making layer. The layer uses Python and TraCI to collect real-time traffic data from SUMO and control its execution.

The Intelligence Layer provides the Reinforcement Learning Engine part which was implemented using PyTorch, responsible for observing the current traffic situation, deciding signal actions, calculating rewards, and learning an optimal policy for adaptive signal control. In addition, the aim of the Traffic Signal Optimization process is to reduce congestion, waiting time, enhance traffic efficiency as well as prioritize emergency and pedestrian movement if necessary.

The Database Layer has SQLite as a lightweight local database used for storing and managing the system's data. Various information is stored in this database including user accounts, traffic scenarios, road limitations, signal control methods, running simulations, traffic indicators, and rewards output. This makes it possible for the system to manage the local information without needing a central database server or cloud information storage.

The Outputs Layer represents the actual outputs of the system which consist of real-time traffic data in the simulation, a queue length and waiting-time analysis, traffic reports, a comparison of reinforcement learning to static time signal control and any dashboard visualizations. The user can analyze the efficiency of the SMARTFLOW system for varying traffic conditions and road limitations through the available outputs.

In conclusion, the architecture demonstrates a combination of standalone systems centered around the use of a simulation based intelligent traffic management system where scenarios are set up by the user, the traffic simulation is run using SUMO, communications are done using Python and TraCI, adaptive signal controls are defined by the reinforcement learning engine, the output data is stored in SQLite and the reports and visualizations are provided using dashboard interface.

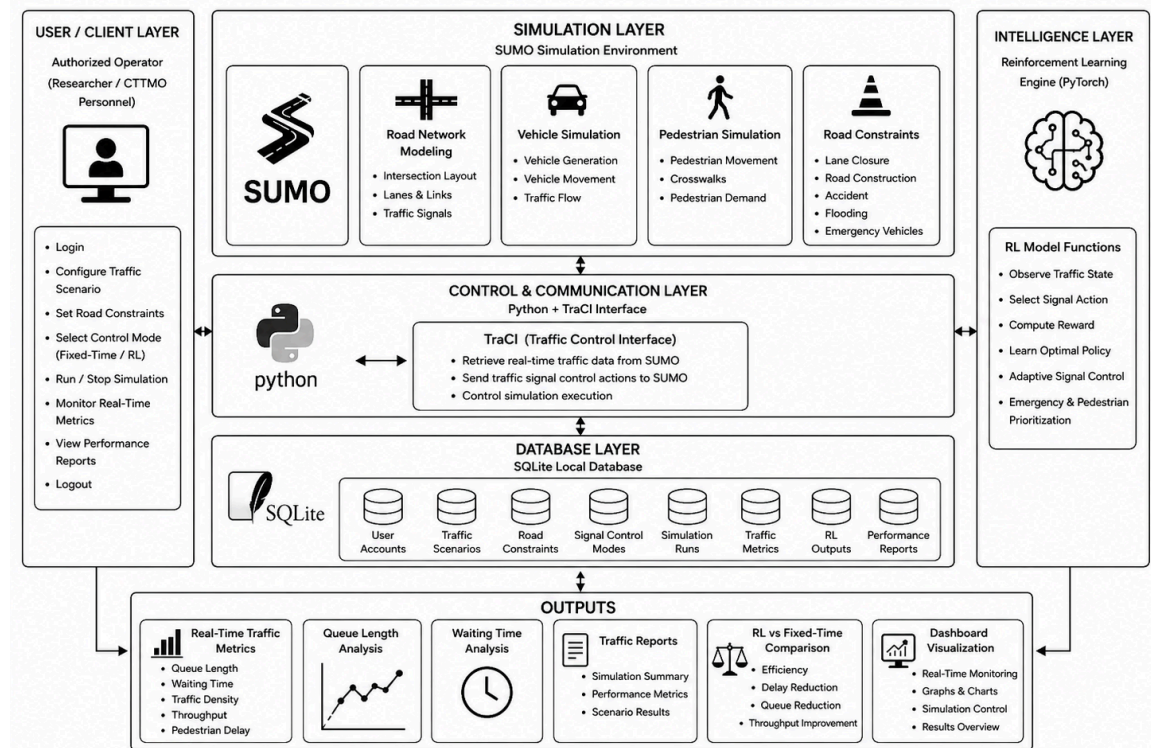


Figure 11. System Architecture Diagram

Conceptual Framework

In Figure 12, the conceptual framework of the SMARTFLOW Traffic System is illustrated. The framework shows how the system obtains configuration inputs from the Authorized Operator, processes this configuration input information through a SUMO-based simulation environment, applies reinforcement learning-based traffic signal control, and generates monitoring results and performance reports using a local dashboard. Based on the requirements of researchers and traffic management personnel, the framework guarantees that the system can facilitate useful tools for configuring traffic scenarios, road constraint conditions, running simulations, monitoring traffic behavior, and comparing reinforcement learning-based signal control with fixed-time traffic signal control.

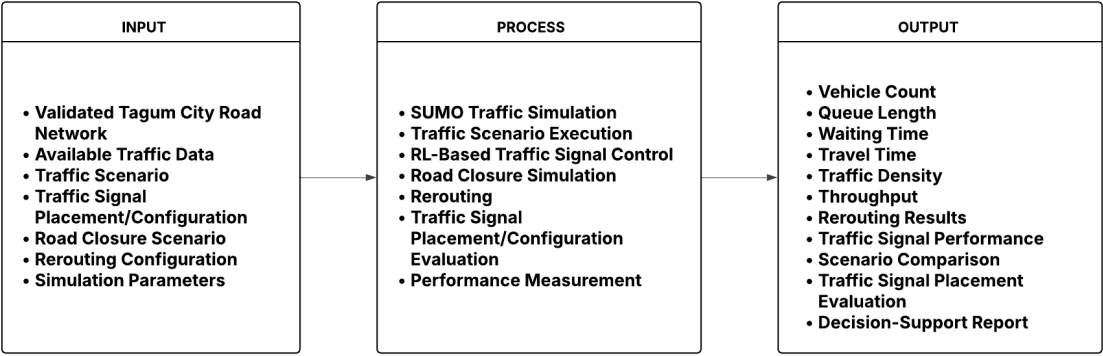


Figure 12. Conceptual Framework

Input

The Authorized Operator will enter the system’s input which involves all configuration settings and data. The input includes login data, traffic scenario configuration data, road constraint configuration data, all simulation commands, and control mode selection. For traffic configuration data, there is traffic density, pedestrian density, and emergency vehicles' status. The road constraint

configuration includes the constraint type, severity level, and the affected lane. The system also receives simulation commands such as initialize, start, pause, stop, reset, and reconfigure simulation. In addition, the operator will select either fixed-time traffic control or reinforcement learning-based traffic control. Each of the inputs will be managed and stored through the local SQLite database before execution.

Process

In the beginning, the Authorized Operator logs into the system via the local dashboard. After signing in, the operator sets up the parameters of the traffic scenario and road constraints. Thus, the system initiates the SUMO simulation environment based on the traffic density configuration, pedestrian density, emergency vehicle status, type of constraint, severity of the constraint, and lane affected. Now during the simulation execution, the communication link between SUMO and signal control units in the system is realized using Python and TraCI. The system collects traffic data from SUMO simulation results (queue length, waiting time, number of vehicles, number of pedestrians, traffic density, and signal state). If fixed-time control is used, the system automatically utilizes predefined signal timing to construct a baseline. If RL-based control is applied, the system evaluates current simulation traffic data, determines adequate signal actions, calculates rewards, and revises the decision regarding signals to enhance traffic flow. The traffic data, signal actions, the output from RL, and simulation logs are all saved in an SQLite database for further analysis.

Output

The system generates outputs which include a local interface that gives traffic monitoring results and performance report based on the simulated data. The output will be able to indicate details about simulation status, traffic parameter, queue length, delay, throughput, congestion level, pedestrian activity, service of emergency vehicles, the state of constraint on road, and signal states.

The output is also able to provide information regarding the reports on performance and comparison of the performance of the model with the reinforcement learning model to that of the fixed-time control based on traffic simulation.

Functional and Non-functional Requirements

Functional and non-functional requirements basically describe and define the required functions and properties of the system expected from the SMARTFLOW Traffic System. Functional requirements address the core functionalities of a system in terms of traffic simulation management including management of simulated traffic and road constraint scenarios, reinforcement learning based operations of traffic signals, and effective performance evaluation of the overall system. Non-functional requirements represent the operational quality based on certain points including efficiency, reliability, usability, maintainability, and adaptability as per ISO/IEC 25010 standards.

Functional Requirements

This system will have the following functions:

1. The system will allow the user to set factors involved in the traffic scenario such as the traffic density, pedestrian density, and status of emergency vehicles.
2. The system will also enable the user to configure constraints for the road, as per the type of constraint, type of traffic regulation, and severity of that type of regulation.
3. The system will simulate the traffic using SUMO (Simulation of Urban Mobility).
4. The system will simulate the movements, as well as interactions between different vehicles and pedestrians.

5. The system will also collect traffic data from the simulation and store it in the database while making use of Python programming language.
6. The system will make use of reinforcement learning to maintain control signals.
7. The system will be able to get signals regarding the phases of the signal based on the data generated from the traffic simulation and will optimize traffic signals as per each situation.
8. The system will keep updating the traffic situation somewhere making use of the previous data regarding the reinforcement process.
9. The system will provide the user with all necessary information in the form of the system dashboard interface.
10. The system will allow the user to start, pause, stop, modify, and reset the traffic simulation with the help of a dashboard interface.
11. The system will save the user credentials, traffic data, and other related information using SQLite (for database), along with an option to export the information into either a CSV or JSON file.
12. Finally, the system will generate final reports based on the different data points collected from the traffic signal optimization and reinforcement method.

Non-Functional Requirements

These are the system non-functional requirements:

1. Simulation scenarios should run and traffic data should be processed in a reasonable time given the complexity of the chosen scenario and road network.
2. Dashboard should be clear and effective enabling authorized users to configure different scenarios, run the simulations, examine results, and download performance reports.
3. The system should be performing efficiently in simulating and it should log the information without any unexpected loss of information. The system

must work with given and confirmed traffic data of the particular road network.

4. The system designed on modular basis; traffic data and traffic simulation can be modified as per need whenever required.
5. The system should support operating systems like SUMO, Python, TraCI, PyTorch, and local database.
6. The system must provide consistency as well as completeness in storage of traffic data and configuration.
7. Configuration and simulation functionalities must be secured from unauthorized access and managed by authentication procedure.
8. It should provide a way to include new road and traffic simulation without any major redesign.
9. It is important for users to perform all the simulations again and again without changing input values; users would be able to make valid comparisons.

Use Case Diagram

The use case diagram presented in Figure 13 illustrates the interaction between an primary actor and the SMARTFLOW Traffic System. The SMARTFLOW Traffic System is a localized simulation-based traffic analysis and decision-support system that authorized operators access to create traffic scenarios and perform testing and reinforcement learning algorithms for traffic signal control.

The primary actor identified in the system is the Authorized Operator, who is responsible for the configuration, control, monitoring, and evaluation of the simulation environment from the local dashboard interface. The dashboard also serves as the primary method of access, allowing the operator to manage simulation activities and analyze how traffic behaves under different configurations.

Primary use cases within the system include user authentication, traffic scenario configuration, road constraint configuration, simulation management, traffic monitoring in the simulation, and system performance evaluation. The authentication portion of the system allows the authorized actor to enter the credentials that are managed locally to access the system.

Traffic scenario configuration enables the Authorized Operator to configure traffic characteristics by setting parameters like the level of traffic density, pedestrian density, and availability of emergency vehicles. The subfunctionality of road constraint configuration allows the operator to define road restrictions by stating the type and level of restriction. These defined conditions help in shaping up the behavior of both vehicles and pedestrians and in adjusting the simulated scenario of operation under MAP.

When the simulation process starts, the system starts using the traffic environment and executing the traffic conditions based on the selected mode of operation in Reverse Engineers. The operator can select the fixed-time control or reinforcement learning control. In case of the reinforcement control mode selection, the RE will provide the adaptive signal decisions dynamically for efficient traffic flow and avoiding congestion.

Via the dashboard interface, the Authorized Operator is capable of observing the traffic status and evaluating the system's operating characteristics throughout the simulation. The outputs of the simulation are displayed in this system, such as the lengths of the queues, average delay, traffic throughput, the states of traffic signals and congestion levels, and pedestrian behavior. The Authorized Operator can also activate, pause, stop, and reset the simulation.

The use case diagram exhibits how authorized users interact with the SMARTFLOW Traffic System to specify traffic conditions, manage driving conditions, operate the execution of the simulation, observe traffic behavior, and evaluate the effectiveness of reinforcement learning–based traffic light planning as opposed to fixed-time planning in the simulation.

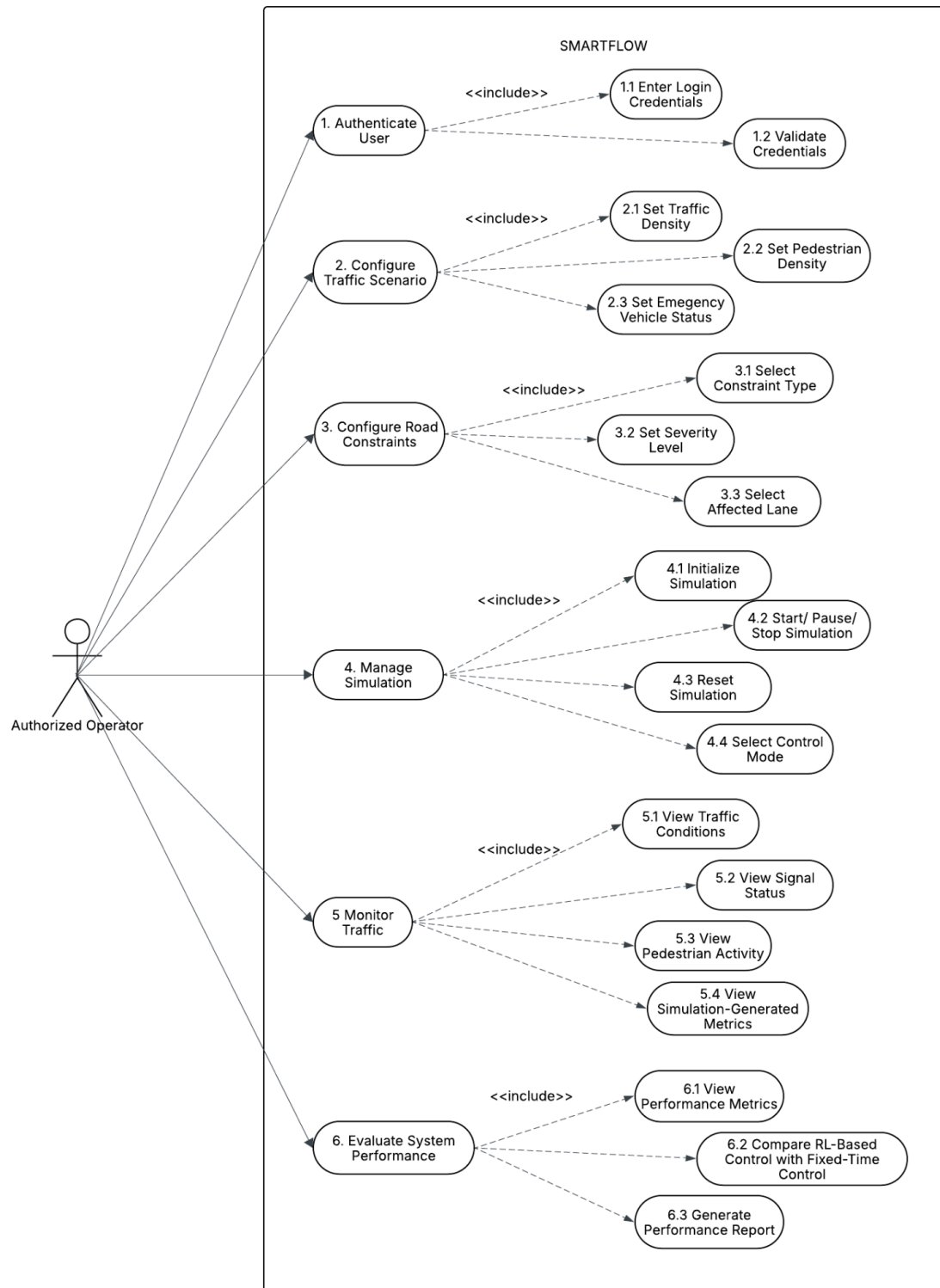


Figure 13. Use Case Diagram

Context Flow Diagram

The Context Flow Diagram as illustrated in Figure 14 shows the whole approach of SMARTFLOW Traffic System and how it interacts with the outside entity and transfer information through simulation-based environment. It highlights how the logical flow of system inputs and outputs without going into internal details of the system.

The important external entity in this diagram is the Authorized Operator that includes all those who might include, people doing research, traffic control personnel and CTTMO employees. The Authorized operator provides input to the system in terms of login details, traffic environment settings, road settings, simulation commands and fixed or reinforcement control modes. The traffic scenario settings include inputs like traffic density, pedestrian density and the status of emergency vehicle; whereas, the road settings contain the type of constraint, severity level and lane details. Based on the settings mentioned above, the system is enabled to initialize through simulation mode accordingly.

The SMARTFLOW Traffic System processes the provided data via traffic simulation, vehicle and pedestrian movement simulation, road constraints implementation, and signal control based on the chosen control mode. In case of reinforcement learning mode, the system uses the trained RL controller for determining the adaptive signal actions while in case of fixed-time mode; the defined traffic signal timing is used in the system for the baseline comparison.

On the constant and interactive dashboard interface, the key outputs generated by the system for the Authorized Operator include real-time vehicular metrics and status, signals statuses, queue lengths, delays, throughput, and any congestion levels across the operational layout along with pedestrians' activities. Thus, the presented results may facilitate making an assessment and comparison of a learning-based adaptive traffic signal control approach with

fixed-time traffic signal control based on the various operational configurations and vehicle flows of the road environment.

As seen in the figure, there are also other important flows of data which are user input, traffic scenarios configuration, road constraints configuration, simulation commands, traffic performances metrics, and reports generation that give an overview of how the SMARTFLOW system processes the incoming input data and converts it into outputs that play an important role in traffic analysis, decision making and simulation based evaluation.

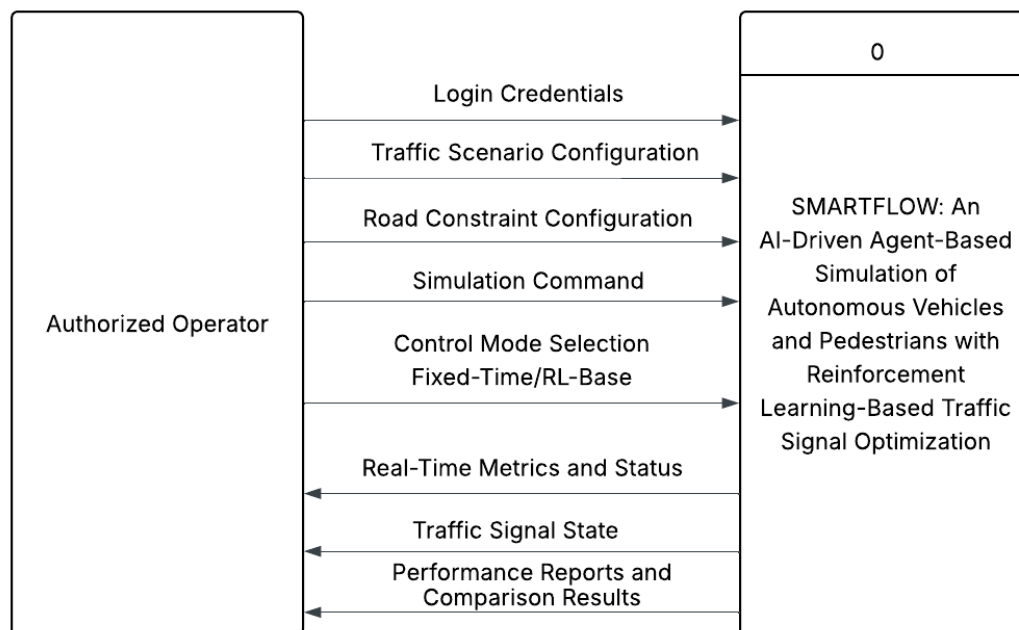


Figure 14. Context flow diagram of the system.

Data Flow Diagram

The Data Flow Diagram illustrated in Figure 15 places the system in five different conceptual processes, namely User Account Authentication, Traffic Scenario and Road Constraint Configuration, Simulation and Data Collection,

Signal Control Processing, and Monitoring and Reporting. The main external entity interacting with the system would be the Authorized Operator, which can be either a researcher or CTTMO personnel. The Authorized Operator will give the login credentials to access the system in addition to providing configurations relating to traffic scenarios, road constraints, simulation commands, and control modes.

The User Account Authentication process verifies the login information provided by the Authorized Operator by sending a user query to the SQLite database and retrieving user account records. After checking for the login credentials, the system will only allow access to the system for authorized personnel making the process very secure.

After gaining authorization, the process of Traffic Scenario and Road Constraint Configuration has been carried out for the management of the inputs of traffic scenarios and road constraint by the Authorized Operator. In the Traffic scenario configuration the various configurations such as density of traffic, the density of pedestrians and the state of emergency vehicles have been made. On the other hand the configuration of the road constraint deals with the types of constraints, severity level of the enforcement and lane affected. The scenarios configurations and the road constraints configured are saved in the SQLite database and the saved scenarios can be retrieved from the database in order to carry out the required simulation.

The Simulation and data collection process gets the commands regarding simulation execution from the Authorized Operator and there are various types of commands executed regarding simulation execution such as starting, pausing, stopping, resetting and running of simulation. This process helps in retrieving the Traffic scenario configurations, road constraint configurations and the simulation parameters from SQLite database. Based on these catalogues the SUMO based traffic simulation is executed and traffic information such as pedestrian activity, vehicle count, queue length, waiting time traffic density and flow of traffic is

collected. The information collected is saved again in the database for its processing purposes.

The Clearing Process refers to the Handling of identified or reported deficiencies in Light Emitting Diode (LED) technology on the Airport Surface Detection Equipment Model 3 (ASDE-3) System. The Deficiency Processing System provides a process flow with these actions and associated text through Structured Query Language (SQL) programming commands. This process provides the ability to monitor, record, edit, and manage deficiencies through updating the accounting activity in the deferral and reporting processes.

The Deficiency Processing System captures the process flow, with each deficiency being received from the airport through SQL programming commands. The process then identifies the deficiency through the action taken which may include reviewing the database for the last time the deficiency was reported through creating an entry in the Deficiency Processing Database via SQL programming commands.

A majority of the deficiencies reported to the FAA involves the ASDE-3 systems receiving units where information is required to validate and quickly identify the deficiency. It also provides the ability to record the data which enables quick validation as well as allows for internal and external reports through the use of the SQL programming commands.

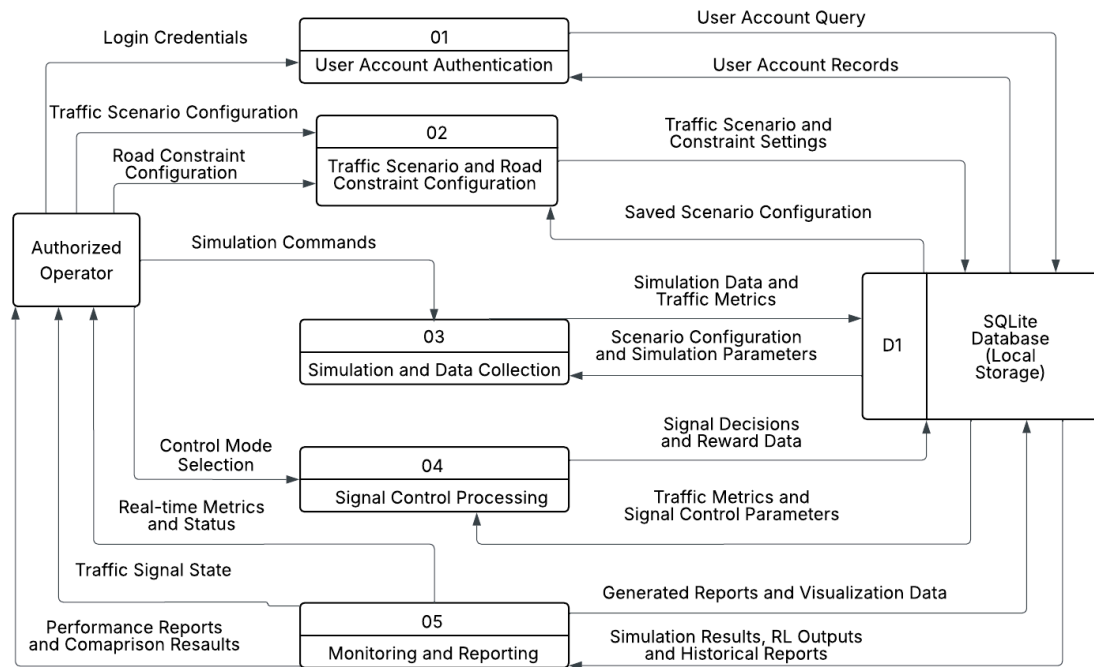


Figure 15. Data flow diagram of the system.

SYSTEM DESIGN

Entity Relationship Diagram

The Entity Relationship Diagram represented in Figure 16 shows the database structure of the SMARTFLOW Traffic System with SQLite being utilized as a light weight database for local systems. Being a localized simulation platform, SQLite is selected for the storage and management of vital system data without the use of centralized databases or cloud storage. The database provides organization for user credentials, traffic scenarios, road constraints, signal durations, simulation runs, traffic data, reinforcement learning data, and reports.

The User Accounts entity has login credentials for users accessing the system's dashboard. Since the system is a localized prototype, it does not have role based access control at this time, and so the account will only be used for

authentication purposes. The Traffic Scenarios entity has the default traffic conditions including normal traffic, heavy traffic, pedestrian heavy conditions and emergency situations. The Road Constraints entity has the disruption values including lane closure, road works, accidents, and pedestrian congestion which can be used in experimental runs.

An entity known as Signal Control Modes indicates how control was applied in each simulation run such as dependent signal control or adaptive control utilizing reinforcement learning. The Simulation Runs entity acts as the main table of the database in order to keep track of the details of each simulation that has taken place and the various parameters that are selected by the user along with the highway condition and mode of control of the signal. Key information regarding each simulation including the commencement time, end time, and current status of execution is included in this entity.

The Traffic Metrics entity has the role to record the summary of the outcomes of the simulations performed. For example queue length, throughput, waiting time, traffic density, and signal efficiency are made available here. Since there is only once outcome generated for each simulation in which traffic metrics is derived, there is a one-to-one relationship between the above mentioned two entities. The Performance Reports are designed to store the summary of performance results based on the execution of each simulation run. Again, there is only one performance report generated from each simulation run thus entailing a one-to-one relationship. At the same time, RL Outputs of the entity will store information relating to various aspects of the model including traffic state, reward value, signal operations made, and timestamps which can result in multiple RL outputs being produced in one simulation run only.

The relationships between the entities of the system have been established by making use of Crow's Foot Notation. Under this framework, one user account can be associated with several simulations while each simulation account will refer to only one user account. Likewise, each traffic scenario, road

restriction, and signal control mode can appear in several simulation runs but each simulation run will refer to only one selection from each of the above. Through a singular simulation, traffic metric and RL outputs can be produced providing possibilities for generating performance reports.

Thus the ERD identifies the way in which SMARTFLOW integrates and organizes its data for performing simulations, carrying out the reinforcement learning process, and conducting performance evaluations. The structure of the database allows the SMARTFLOW program to be able to store data on execution of simulations in the local database and keep record files maintained properly and execute comparison between the use of adaptive control and fixed time traffic control modes.

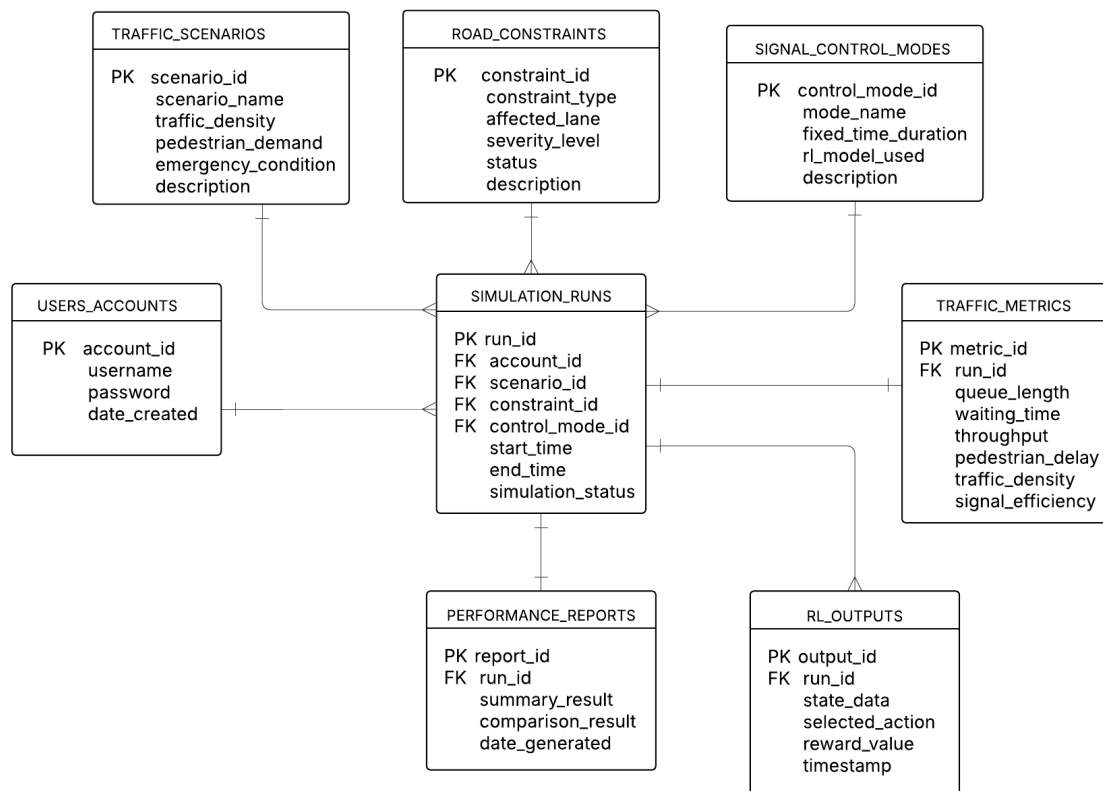


Figure 16. Entity Relationship Diagram

Data Dictionary

The SMARTFLOW Traffic System Data Dictionary gives an extensive overview of the elements of data that make up the SQLite local database used in the system. This aids in comprehending how both the storage and arrangement of data occur while using functions such as user authentication, configuring traffic scenarios, managing road constraints, performing simulations, tracking the output from reinforcement learning, and generating performance reports.

The database is made up of many tables some of which include Users, Traffic Scenarios, Road Constraints, Signal Control Modes, Simulation Runs, Traffic Metrics, RL Outputs, and Performance Reports. Each of these tables has columns or fields that signify the specific type of data a given table contains as well as its necessary conditions and descriptions for effective data entry and retrieval related to the simulation.

Table 2. User

Field Name	Data Type	Length	Constraints	Description
account_id	INTEGER	-	PRIMARY KEY	Unique identifier for each authorized user
username	TEXT	50	UNIQUE, NOT NULL	Username used for local system login
password	TEXT	255	NOT NULL	Password used for authentication
date_created	DATETIME	-	NOT NULL	Date and time the user record was created

Table 3. Traffic Scenarios

Field Name	Data Type	Length	Constraints	Description
scenario_id	INTEGER	-	PRIMARY KEY	Unique identifier for each traffic scenario
scenario_name	TEXT	100	NOT NULL	Name of the traffic scenario
traffic_density	TEXT	30	NOT NULL	Traffic density classification such as normal or heavy
pedestrian_demand	TEXT	30	NOT NULL	Level of pedestrian activity in the simulation
emergency_condition	TEXT	30	NULL	Indicates presence of emergency vehicle conditions
description	TEXT	255	NULL	Additional description of the scenario

Table 4. Road Constraints

Field Name	Data Type	Length	Constraints	Description
constraint_id	INTEGER	-	PRIMARY KEY	Unique identifier for each road constraint

constraint_type	TEXT	50	NOT NULL	Type of road constraint such as lane closure or accident
affected_lane	TEXT	50	NOT NULL	Lane or road section affected by the constraint
severity_level	TEXT	20	NOT NULL	Severity classification of the constraint
status	TEXT	20	NOT NULL	Current status of the road constraint
description	TEXT	255	NULL	Additional details regarding the constraint

Table 5. Signal Control Modes

Field Name	Data Type	Length	Constraints	Description
control_mode_id	INTEGER	-	PRIMARY KEY	Unique identifier for each control mode
mode_name	TEXT	50	NOT NULL	Name of the signal control mode
fixed_time_duration	INTEGER	-	NULL	Preset signal duration for fixed-time control
rl_model_used	TEXT	100	NULL	Name or identifier of the RL model used
description	TEXT	255	NULL	Additional information about the control mode

Table 6. Simulation Runs

Field Name	Data Type	Length	Constraints	Description
run_id	INTEGER	-	PRIMARY KEY	Unique identifier for each simulation run
account_id	INTEGER	-	FOREIGN KEY	Reference to the authorized user
scenario_id	INTEGER	-	FOREIGN KEY	Reference to the selected traffic scenario
constraint_id	INTEGER	-	FOREIGN KEY	Reference to the applied road constraint
control_mode_id	INTEGER	-	FOREIGN KEY	Reference to the selected signal control mode
start_time	DATETIME	-	NOT NULL	Simulation start timestamp
end_time	DATETIME	-	NULL	Simulation end timestamp
simulation_status	TEXT	30	NOT NULL	Current status of the simulation

Table 7. Traffic Metrics

Field Name	Data Type	Length	Constraints	Description
metric_id	INTEGER	-	PRIMARY KEY	Unique identifier for traffic metric data

run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run
queue_length	REAL	-	NOT NULL	Measured vehicle queue length
waiting_time	REAL	-	NOT NULL	Average waiting time during simulation
throughput	REAL	-	NOT NULL	Number of vehicles passing through the intersection
pedestrian_delay	REAL	-	NULL	Delay experienced by pedestrians
traffic_density	REAL	-	NULL	Measured traffic density
signal_efficiency	REAL	-	NULL	Efficiency score of signal control

Table 8. RL Outputs

Field Name	Data Type	Length	Constraints	Description
output_id	INTEGER	-	PRIMARY KEY	Unique identifier for RL output data
run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run

state_data	TEXT	500	NOT NULL	Encoded traffic state information
selected_action	TEXT	100	NOT NULL	Traffic signal action selected by the RL model
reward_value	REAL	-	NOT NULL	Reward value generated during RL processing
timestamp	DATETIME	-	NOT NULL	Date and time the RL action was generated

Table 9. Performance Reports

Field Name	Data Type	Length	Constraints	Description
report_id	INTEGER	-	PRIMARY KEY	Unique identifier for each performance report
run_id	INTEGER	-	FOREIGN KEY	Reference to the simulation run
summary_result	TEXT	1000	NOT NULL	Summary of simulation performance results
comparison_result	TEXT	1000	NULL	Comparison between fixed-time and RL-based dual control
generated_by	TEXT	100	NULL	User or process that generated the report

date_generated	DATETIME	-	NOT NULL	Date and time the report was generated
----------------	----------	---	----------	--

Technologies, Concepts, and Theories

In this section, the essential models, algorithms, concepts, and technologies employed in the development of the SMARTFLOW Traffic System are discussed. The manner in which traffic simulation data is produced, processed, analyzed, and what is the use of this information for optimizing traffic signals.

Data Collection

Data collection in SMARTFLOW takes place in the SUMO simulation environment. This means that SMARTFLOW does not rely on live CCTV feeds, IoT sensors, or GPS devices, instead, it collects data related to traffic generated during simulation runs. The available information consists of vehicle count, pedestrian count, queue length, waiting time, traffic density, traffic volume, pedestrian delay, presence of emergency vehicles, and traffic signal status. Using TraCI, the simulation values are accessed in real-time from Python and sent to the control and decision-making parts of the system.

Data Pre-Processing

Before the training data is fed into the reinforcement learning controller, it is pre-processed first. The main goal of the pre-processing is to make sure that the data is formatted according to the required specifications, organized properly, and is ready for analysis and evaluation.

Feature Extraction

The process of feature extraction is essentially that of identifying the most relevant attributes of the traffic that will be used in the decision to control signals.

The features selected in SMARTFLOW may include queue length in each lane, average wait time, vehicle density, number of pedestrians, emergency vehicles, signal phase, and road fall-back status meaning the attributes in question correspond to the current traffic situation at the simulated intersection representing the input state for the reinforcement learning model.

Reinforcement Learning-Based Decision-Making

The Decision-Making process at the SMARTFLOW Traffic System is conducted using Reinforcement Learning for the adaptive control of traffic signals. In SMARTFLOW, the learning agent is the traffic signal control system and the traffic environment is represented by the simulated system allowing observations of traffic situations and corresponding signal actions through TraCI.

The agent will monitor the current state of traffic in the intersection such as queue length, number of vehicles, average time of waiting, traffic density, the need for pedestrians, presence of emergency vehicles, the state of the road and current phase of the signal at the intersection. After observing all these characteristics, the agent will decide what action has to be taken regarding the signals e.g. whether to shift to the next appropriate phase or for instance, prolong or shorten the current phase according to the traffic situation at hand.

It is important to note that instead of a fixed timing of the signals, the agent will learn from various traffic situations and find out whether its own action has improved traffic conditions. Significant aspects of traffic performance observed by the agent include waiting time of vehicles, length of the queue, amount of pedestrian traffic, traffic flow and priority of emergency vehicles.

Reinforcement learning is done using simulated data instead of any live data such as CCTV, GPS, or physical sensors. Each simulation is regarded as a separate episode thus providing different conditions regarding traffic characteristics in order to evaluate whether the action results in improved traffic conditions or not.

In SMARTFLOW, input traffic characteristics are generated during simulations and then transferred through TraCI so that the agent can decide upon a corresponding action and results from the previous actions may support better decision making in the future.

In the present study, the reinforcement learning-based approach will be validated by means of comparing results with a fixed-time signal control approach so that further improvements such as shorter waiting times, shorter queues, better efficiency and priority for emergency vehicles, etc., may be achieved.

Technologies Used in the System

Simulation of Urban Mobility (SUMO)

The system utilizes mainly Simulation of Urban Mobility (SUMO), an open-source platform for traffic simulation that allows for roadway designs to support simulation of traffic flows in vehicular movement and pedestrian activities. It serves as the main simulation environment of the SMARTFLOW system, thus providing the developers of SMARTFLOW with an opportunity to design realistic traffic conditions, street constraint conditions, and traffic flows within the intersection of Tagum City under study. Because of SUMO, the current system can simulate several traffic densities, pedestrian activities, emergency vehicle response, traffic congestion during road closures, among many other things.

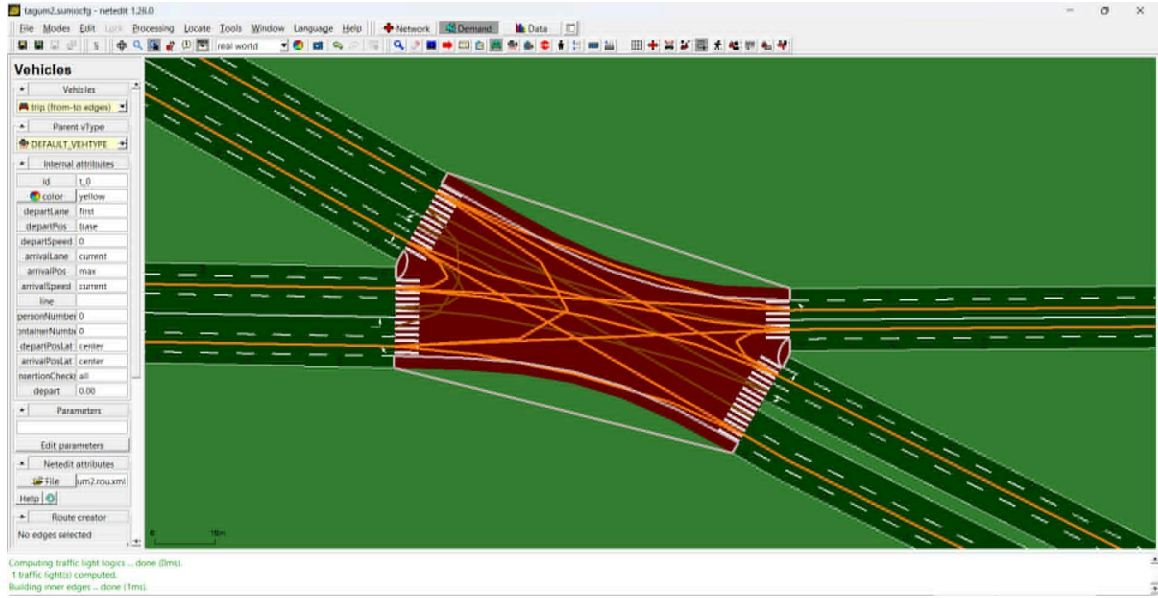


Figure 17. Simulation of Urban Mobility (SUMO)

Reinforcement Learning (RL)

Because it enables adaptive traffic signal optimization, the system employs Reinforcement Learning (RL) as part of its technique. Due to the RL technique, the system can determine the traffic signal control mechanism through continued interaction with the simulation system. The RL model observes the traffic conditions, executes the control strategy and receives a reward. As a result, the modelling methodology helps improve the decision-making capabilities of the SMARTFLOW system.

```

1 from __future__ import annotations
2
3 import json
4 import statistics
5 from dataclasses import asdict, dataclass
6 from datetime import UTC, datetime
7 from pathlib import Path
8 from typing import Callable
9
10 try:
11     from stable_baselines3 import DQN
12     from stable_baselines3.common.callbacks import BaseCallback
13 except ImportError: # pragma: no cover - exercised when the local venv lacks the RL stack
14     DQN = None
15     BaseCallback = None
16
17
18 OBSERVATION_VERSION = "obs_v1"
19 REWARD_VERSION = "reward_v1"
20 ACTION_SPACE_VERSION = "action_v1"
21
22
23 @dataclass(frozen=True)
24 class DQITrainingConfig:
25     total_timesteps: int
26     learning_rate: float = 1e-4
27     buffer_size: int = 50_000
28     learning_starts: int = 100
29     batch_size: int = 32
30     gamma: float = 0.99
31     train_freq: int = 1
32     gradient_steps: int = 1
33     target_update_interval: int = 250
34     exploration_fraction: float = 0.2
35     exploration_initial_eps: float = 1.0
36     exploration_final_eps: float = 0.05
37     seed: int = 0

```

Figure 18. Reinforcement Learning (RL)

Python and TraCI

Python serves as the programming language that connects the simulation environment with the reinforcement learning model through the dashboard functionality and databases of the simulation system. Because of its modular nature, it provides an organized way of developing and integrating the parts of the system. An extremely important Python module for traffic simulation is the Traffic Control Interface (TraCI), which enables software access to the running simulation in SUMO and provides Python software modules with the traffic control command.

```

111 class SumoSimulationEngine:
234     def configure(self, **kwargs):
245         if kwargs:
247             def configure_from_scenario(self, scenario: dict):
248                 if scenario.get('intersection_id'):
249                     self.set_intersection(scenario['intersection_id'])
250                 if scenario.get('name'):
251                     self.current_scenario_name = scenario['name']
252                 self.configure(**{
253                     key: scenario[key]
254                     for key in (
255                         "traffic_density",
256                         "pedestrian_density",
257                         "emergency_mode",
258                         "road_constraint",
259                         "lane_closure",
260                         "construction",
261                         "accident",
262                         "flooding",
263                         "temp_blockage",
264                     )
265                     if key in scenario and scenario[key] is not None
266                 })
267             def configure_rl_control(
268                 self,
269                 ",
270                 decision_interval: float = 5.0,
271                 minimum_green_hold: float = 10.0,
272                 controller_label: str = "RL Controller",
273                 controller_provenance: str = "rl",
274             ):
275                 self.rl_control_enabled = True
276                 self.rl_decision_interval = max(float(decision_interval or 5.0), SUMO_STEP_LENGTH)
277                 self.rl_minimum_green_hold = max(float(minimum_green_hold or 10.0), SUMO_STEP_LENGTH)
278                 self.rl_pending_action = None

```

Figure 19. Python and TraCI

PyTorch

The Deep Q-Network (DQN) model to be implemented in SMARTFLOW Traffic System's reinforcement learning module will use PyTorch as the software framework. PyTorch is well-acclaimed for its optimized tensor manipulation and deep learning advantages that are essential for the training of neural network-based models on both CPUs and GPUs. Its applicability in SMARTFLOW Traffic System comes from its utilization in constructing DQN model implementation that estimates optimal traffic signal operation from all attributes concerning the present state of simulation which may comprise, but is not limited to queue length, waiting time, traffic density, pedestrian requirement, presence of emergency vehicle, and road conditions. It will also play a key role in training the model through simulation experiences generated in SUMO episodes, the calculation of training loss, and adjustment of model parameters for improved adaptive traffic signal decision-making.

```

C:\Users\jhepo\Desktop> trapik2 > simulation > rl.policy.runtime.py > extract_dql_q_values
89 def masked_argmax(values: Iterable[float], valid_action_mask: tuple[int, ...] | None = None) -> int:
96     for index, value in enumerate(q_values):
97         if valid_action_mask is not None and (index >= len(valid_action_mask) or not valid_action_mask[index]):
98             continue
99         if best_value is None or value > best_value:
100             best_action = index
101             best_value = value
102     if best_action is None:
103         return first valid action(valid_action_mask)
104     return int(best_action)
105
106
107 def _extract_dql_q_values(model, observation) -> tuple[float, ...]:
108     if hasattr(model, "predict_q_values"):
109         return tuple(float(value) for value in model.predict_q_values(observation))
110
111     try:
112         import numpy as np
113         import torch
114     except ImportError:
115         return ()
116
117     if observation is None or not hasattr(model, "q_net"):
118         return ()
119
120     try:
121         observation_array = np.asarray(observation, dtype=np.float32)
122         if observation_array.ndim == 1:
123             observation_array = observation_array.reshape(1, -1)
124         observation_tensor = torch.as_tensor(observation_array, device=model.device)
125         with torch.no_grad():
126             q_values = model.q_net(observation_tensor)
127         return tuple(float(value) for value in q_values.detach().cpu().numpy()[0])
128     except Exception:
129         return ()
130

```

Figure 20. PyTorch

Database Management System

To handle data in SMARTFLOW Traffic System, the users can take advantage of SQLite as the DBMS for storing and organizing user data, traffic cases, road constraints, simulation results, reinforcement learning outcomes, and performance reports in order to allow ease of retrieval of all relevant information as pertains simulation of such processes. The database can also store data locally without the need for internet connection or the use of a centralized database server to make the system to be lightweight and reliable.

A screenshot of a code editor window titled 'database.py M'. The editor shows Python code for a database management system. The code includes imports for sqlite3, os, json, secrets, datetime, UTC, timedelta, contextlib, contextmanager, and pathlib. It defines a list of scenario JSON fields and a function to get a database connection. The code is as follows:

```
1 """
2 SMARTFLOW - SQLite Database Layer
3 All schema definitions, CRUD operations, and seed data.
4 """
5
6 import sqlite3
7 import os
8 import json
9 import secrets
10 from datetime import UTC, datetime, timedelta
11 from contextlib import contextmanager
12 from pathlib import Path
13
14 import config
15
16 SCENARIO_JSON_FIELDS = (
17     "lane_closure_config",
18     "construction_config",
19     "accident_config",
20     "flooding_config",
21 )
22 AUTH_SESSION_VERSION_SETTING = "auth_session_version"
23
24
25 def get_connection():
26     os.makedirs(os.path.dirname(config.DB_PATH), exist_ok=True)
27     conn = sqlite3.connect(config.DB_PATH)
28     conn.row_factory = sqlite3.Row
29     conn.execute("PRAGMA journal_mode=WAL")
30     conn.execute("PRAGMA foreign_keys=ON")
31     conn.execute("PRAGMA busy_timeout=5000")
32     return conn
33
34
35 def _connect_sqlite_file(path: str | Path) -> sqlite3.Connection:
36     conn = sqlite3.connect(str(path))
37     conn.row_factory = sqlite3.Row
```

Figure 21. Database Management System (DBMS)

SYSTEM TESTING AND IMPLEMENTATION

The System Testing and Implementation phase is concerned with confirming the operational efficiency of the SMARTFLOW Traffic System by validating the system requirements to be met. In this phase, extensive examination of the entire program will be performed to ascertain that the major components of the system are successfully integrated into one another. This integration process unfolds as a testing process where the first examination is undertaken to check how well each part of the SMARTFLOW system performs through various metrics applicable to the system. One of the key aspects of testing revolves around matching the system specifications regarding the waiting time, the number of vehicles queued up, traffic density, and efficiency of the traffic control signals.

Thus, the system will undergo unit testing, integration testing, and scenario-application testing so as to be assured of the successful operation of the connection. The test cases will be developed in a manner that they will enable the simulation of both regular and disruptive transport modes in order to

check whether the system will be able to interpret, process, and produce relevant data in a stable manner.

Apart from functional testing, the RL-based traffic signal controller will also be subjected to tests conducted by using the RL-related performance indicators, like reward, rate of convergence, stability of the policy, average waiting time, the length of the queue, etc. These measurements will be taken into account in order to assess how efficiently and effectively the model with RL algorithms can process the traffic signal requests and establish proper control over the vehicles and different mobile devices.

A testing results summary presented in tabular format shows the operating status and performance of each major component, such as traffic simulation, reinforcement learning process, signal control operations, database storage, dashboards, and report performances. This will enable the proponents to monitor issues and ensure that the functions of the SMARTFLOW system meet the required operational standards.

In order to facilitate the validations process, a System Test Plan should be drafted that defines the goals, scope, environment, test cases, procedures, and acceptance criteria for testing. This enables testing to take place in an organized, orderly, and systematic fashion according to defined functional and non-functional requirements.

The implementation phase occurs when the SMARTFLOW system is deployed in simulated desktop local environment. The implementation includes configuring the SUMO platform, establishing Python and TraCI communication, developing the reinforcement learning algorithm, setting up SQLite database, and building the local dashboard. The SMARTFLOW system will be used by users and pre-authorized traffic managers for testing out traffic scenarios before real implementation. The SMARTFLOW system performs the functions of the Reinforcement Learning based traffic signal controller for validation purposes. Further validations will be conducted to ensure that the SMARTFLOW system

performs in the prescribed environment. Additionally, some persons will be trained on how to use the simulation dashboard and traffic analysis features of SMARTFLOW system.

In conclusion, After simulation, the SMARTFLOW system will be monitored and evaluated for its efficiency in order to improve adaptive traffic signal optimization system further.

SYSTEM MAINTENANCE

The System Maintenance phase is concerned with the maintenance of the SMARTFLOW Traffic System after its deployment. The System Maintenance phase involves the observation of the performance of the system, diagnosing malfunctions in the system, and implementing required modifications based on detected faults that will improve the functional accuracy of the simulation as well as the reinforcement learning quality in the system and enhance the usability of the application.

Some of the maintenance tasks include periodic checks on the SUMO simulator, upgrades to the Python and reinforcement learning components, improvements to the road constraints and traffic scenario setups, enhancements of the graphical user interface and influence over the SQLite database. The implementers may modify the simulation parameters and optimize the reinforcement learning behavior based on test results and users' responses.

In addition to that, constant monitoring of the traffic simulation results as well as the behavior of the adaptive signal control may continue to ensure that future improvements will be possible and the number of possible traffic scenarios can be increased.

System Security Plan

In order to protect the data in the system and resources associated with the simulation, an implementation process has to go through an application of a System Security Plan for the SMARTFLOW Traffic System. The approach adopted in this system shall ensure user authentication and role-based control. The security requirement protects unauthorized access to the system dashboard, database, and the system configuration. The functionality of the system exists locally; therefore, simulation data, results of reinforcement learning as well as configuration data are both owned locally.

The system uses secure communication method and secure processing that is enhanced by stable communication between the SUMO simulation environment, the communication medium associated with Python-TraCI, features of reinforcement learning, an SQLite database that helps reduce vulnerability of the components in the system. The aspects of input validation shall be implemented into the system to ensure proper operation and protection against malicious inputs and scripts.

System Security Plan provided in Table IV implements ISO/IEC 27001-security practices, which will allow protecting confidentiality, integrity, and availability of data in SMARTFLOW Traffic system. The characterized security controls will provide for secure system operations, reliable processing of simulation results, as well as secure management of reinforcement learning outputs.

Table 10. System Security Plan

Security Domain (ISO 27001)	Control Area	Description / Implementation in the System
------------------------------------	---------------------	---

A.5 Information Security Policies	Security Policy	The system will implement security policies for user access, simulation control, and local data protection.
A.6 Organization of Information Security	Roles and Responsibilities	Roles such as Administrator, Researcher, and Authorized Operator will have defined system privileges.
A.9 Access Control	User Access Management	The system will enforce role-based access control (RBAC) through secured login credentials.
A.9 Access Control	Authentication Mechanism	Username and password authentication will be implemented to restrict unauthorized access.
A.10 Cryptography	Data Protection	Local simulation data and exported files will be protected through secure storage and backup mechanisms.
A.12 Operations Security	Data Processing Integrity	Input validation and error handling will ensure accurate traffic simulation processing and stable system operation.

A.12 Operations Security	Logging and Monitoring	The system will maintain logs of simulation activities, user actions, and system events for monitoring and troubleshooting.
A.13 Communications Security	Secure Communication	Communication between SUMO, Python, TraCI, and reinforcement learning components will follow secure processing practices.
A.11 Physical and Environmental Security	Device Protection	Desktop devices and local storage used for simulation execution will be secured from unauthorized physical access.
A.17 Information Security Aspects of Business Continuity	Data Backup and Recovery	Regular backup of SQLite databases and simulation reports will be performed to ensure data recovery.
A.16 Information Security Incident Management	Incident Response	Procedures will be established for detecting, reporting, and resolving system errors or security-related issues.

A.14 System Acquisition, Development, and Maintenance	Secure Development	Secure coding practices, debugging, and system updates will be applied throughout development and maintenance.
---	--------------------	--

System Maintenance Plan

A System Maintenance Program is to be developed to ensure that the continued operation, advancement, and sustainability of the SMARTFLOW Traffic System will be attained once it has been implemented and running. Included in this program are scheduled maintenance operations that involve simulation optimization, reinforcement learning model improvements, SQLite database maintenance, dashboard enhancements, and performance monitoring of the system. Implementation of troubleshooting, debugging, and error correction protocols will also be done to attain the continuous stable performance of the simulation.

In addition, system documentation will be continued to be updated to show necessary information regarding configuration changes, reinforcement learning change, simulation updates, and the performed maintenance. This will ensure that the SMARTFLOW system can be enhanced more easily in the future for various traffic scenarios and intelligent transportation studies.

The System Maintenance Plan shown in Table 11 is developed to link the maintenance operations to the ISO/IEC 25010 software quality attributes which ensure that the SMARTFLOW Traffic System will continue to be reliable, efficient, maintainable, secure, and flexible.

Table 11. System Maintenance Plan

ISO 25010 Attribute	Maintenance Activity	Description	Frequency
Reliability	System Monitoring	Monitoring simulation execution, system stability, and dashboard operation	Continuous
Performance Efficiency	Performance Optimization	Improving simulation processing speed and reinforcement learning performance	Regularly
Usability	Dashboard Improvement	Updating dashboard layout and controls based on user feedback	As Needed
Security	Security Updates	Applying system updates, access control improvements, and database backup procedures	Periodically
Maintainability	Debugging and Code Refactoring	Correcting bugs and improving system structure for easier maintenance	Regularly

Compatibility	System Updates	Ensuring compatibility with SUMO, Python, PyTorch, and desktop environments	As Needed
Reliability	Simulation Calibration	Adjusting traffic simulation parameters and reinforcement learning configurations	Periodically
Performance Efficiency	Database Optimization	Managing SQLite database cleanup, indexing, and report storage optimization	Monthly

REFERENCES:

- [1] Injac, Z., Arsić, S., Drašković, D., & Arsić, M. (2025). Urban traffic management using artificial intelligence: A sustainable approach to enhancing urban mobility. *Journal of Traffic and Transport Theory and Practice (JTTP)*, 10(1), 30-35.
- [2] Akinade, A.O., Adepoju, P.A., & Ige, A.B. Artificial Intelligence in Traffic Management: A Review of Smart Solutions and Urban Impact.
- [3] Rathore, S. P. S., Farhaoui, Y., Aniebonam, E. E., Nagpal, T., & Kaushik, P. (2025, March). AI-driven traffic congestion management: a predictive analytics approach for smart cities. In 2025 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) (Vol. 3, pp. 1-6). IEEE.
- [4] S. Rudhra, S. Vinod, P. Arthi and L. Kurinjimalar, "Density Based Traffic Signal System," 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2024, pp. 1-3, doi: 10.1109/ICPECTS62210.2024.10780421.
- [5] Gheorghe, C., & Șoica, A. (2025). Revolutionizing Urban Mobility: A Systematic Review of AI, IoT, and Predictive Analytics in Adaptive Traffic Control Systems for Road Networks. *Electronics*.
- [6] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [7] Huang, S. Y., Chang, H. C., Chen, Y. C., Wei, T. H., Yeh, I. H., Kuan, S. Y., ... & Wu, I. (2026). A Robust and Efficient Multi-Agent Reinforcement Learning Framework for Traffic Signal Control. *arXiv preprint arXiv:2603.12096*.
- [8] Almomany, A., Eedi, E., & Sutcu, M. (2025). Real-time traffic signal optimization for urban mobility: a reinforcement learning-enhanced framework with application to Kuwait City. *Frontiers in Robotics and AI*, 12, 1669952.
- [9] United Nations, "Goal 9: Build resilient infrastructure, promote sustainable industrialization and foster innovation," *Sustainable Development Goals*, 2023.

- [10] United Nations, "Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable," Sustainable Development Goals, 2023.
- [11] M. Muralidharan and R. Pedarsani, "Analysis of fixed-time control in signalized intersection networks," *IEEE Transactions on Control of Network Systems*, vol. 2, no. 1, pp. 22-31, 2014.
- [12] Sylva, V., & Lala, A. (2026). Microsimulation-Based Evaluation of Fixed-Time and Vehicle-Actuated Traffic Signal Control Strategies Using PTV Vissim in an Urban Corridor. *Informatica*, 50(5).
- [13] C. Aboy, M. Chua, S. Gochuico, J. Nuesca, and M. Suwanpimol, "PC-Based Adaptive Road Traffic Control System," Undergraduate thesis, Dept. Electron. Commun. Eng., De La Salle Univ., Manila, Philippines, 1994.
- [14] Geotab, "Video Intelligence and AI-driven enforcement in the Philippines," *PR Newswire Asia*, May 2026.
- [15] Grand View Research, "Philippines Intelligent Transportation System Market Size & Outlook,"
- [16] DOST-PCIEERD, "An Analysis of Smart City Development Frameworks in the Philippines," *UP CIDS Policy Brief*
- [17] Dubey, A., Lakhani, M., Dave, S., & Patoliya, J. J. (2017, December). Internet of Things based adaptive traffic management system as a part of Intelligent Transportation System (ITS). In 2017 international conference on soft computing and its engineering applications (icSoftComp) (pp. 1-6). IEEE.
- [18] Zemmouchi-Ghomari, L. (2025). Artificial intelligence in intelligent transportation systems. *Journal of Intelligent Manufacturing and Special Equipment*, 6(1), 26-42.
- [19] Damadam, S., Zourbakhsh, M., Javidan, R., & Faroughi, A. (2022). An Intelligent IoT Based Traffic Light Management System: Deep Reinforcement Learning. *Smart Cities*, 5 (4), 1293-1311.
- [20] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [21] H. Zhang, "Multi-Agent Reinforcement Learning Framework for Traffic Signal Control," *arXiv preprint arXiv:2603.12096*, 2026.

- [22] T. Miller, "Quantifying the Impact of RL on Urban Congestion," *Journal of Urban Systems*, vol. 15, no. 4, pp. 201-215, 2025.
- [23] A. Al-Wajih, "AI in ITS: Real-Time Decision-Making Architectures," *IEEE Access*, vol. 13, pp. 14502-14518, 2025.
- [24] M. J. Williams, "Agent-Based Modeling of Urban Mobility Landscapes," *IEEE Transactions on Smart Cities*, vol. 9, no. 2, pp. 112-125, 2024.
- [25] D. Krajzewicz, "SUMO - Simulation of Urban MObility," *IEEE Intelligent Transportation Systems Conference*, 2012.
- [26] X. Ma, X. Hu, and D. Schramm, "Traffic Demand Accuracy Study Based on Public Data," *Applied Sciences*, vol. 15, no. 21, p. 11589, Oct. 2025, doi: 10.3390/app152111589.
- [27] S. Chin, O. Franzese, D. Greene, H. Hwang, and R. Gibson, "Temporary Losses of Highway Capacity and Impacts on Performance," 2002. Accessed: Aug. 11, 2026. [Online]. Available: <https://info.ornl.gov/sites/publications/Files/Pub57166.pdf>
- [28] Osogami, T., Mizuta, H., & Idé, T. (2013, October). Identifying the optimal road closure with simulation. In *Proceedings of the 20th ITS World Congress Tokyo (Vol. 2013)*.
- [29] Rakha, H., & Van Aerde, M. (1995). Statistical analysis of day-to-day variations in real-time traffic flow data. *Transportation research record*, 26-34.
- [30] Ahmad, O. (2005, July). Issues related to the commonality and comparability of driving simulation scenarios. In *IMAGE Conference*.
- [31] Papageorgiou, G., Damianou, P., Pitsillides, A., Ahamis, T., & Ioannou, P. (2007). A computer simulation scenario analysis approach as a decision support tool for transportation systems planning. *WIT Transactions on The Built Environment*, 96, 177-187.
- [32] Kodama, N., Harada, T., & Miyazaki, K. (2022). Traffic signal control system using deep reinforcement learning with emphasis on reinforcing successful experiences. *IEEE Access*, 10, 128943-128950.
- [33] S. Suo, S. Regalado, S. Casas, and R. Urtasun, "TrafficSim: Learning to Simulate Realistic Multi-Agent Behaviors," in *Proceedings of the*

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, pp. 10400-10409.

- [34] K. AlMulla, S. Ashkanani, F. Hassan, H. AlMesbahi, M. Alazmi, and M. Nadeem, "IoT-Based Adaptive Traffic Signal Controller to Optimize the Flow of Traffic and Reduce Congestion," in *Proceedings of the 2024 1st Mediterranean Smart Cities Conference (MSCC)*, 2024